



Working Report

Benchmark WKMSYSPiCT Meeting

Stochastic Surplus Production Model in Continuous Time (SPiCT) Explorations in Nephrops FU30 Stock (Gulf of Cadiz)

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Abstract

Nephrops stock in FU30 Division 9.a is considered a Category 3 stock. The advice is currently based on the 2-over-3 rule using the abundance index from UWTV surveys. The reference points for this stock are undefined and ICES cannot assess the stock and exploitation status relative to MSY or precautionary approach (PA) reference points. Stochastic surplus production model in continuous time (SPiCT) has been recommended by ICES for a number of data limited stocks to produce MSY advice. This WD shows some SPiCT explorations carried out in order to evaluate the potential of this model for *Nephrops* FU 30 assessment and propose this stock as possible candidate for the next WKMSYSPiCT benchmark. Results obtained are promising.

Introduction

Nephrops stock in FU30 Division 9.a was benchmarked in October 2016 ([ICES, 2017](#)) and an approach based on UWTV surveys to generate catch options was proposed for this FU. However, reference points could not be derived according to methodologies used by Category 1 Nephrops stocks conducting UWTV surveys. Consequently, the stock was not upgraded to Category 1. Different assessment models developed for data-limited stocks (DLS), such as Length-Based Indicators (LBIs) or Mean Length-Z (MLZ) based on the WKLIFE V framework ([Acom, 2015](#)), as well as the Separable Cohort Analysis (SCA; R package version 1.2.0; Bell ([2019](#))) and the Separable Length Cohort Analysis (SLCA-nepref; R package version 0.2.2; Dobby ([2019](#))), which are used to calculate MSY reference points for Category 1 Nephrops stocks, were explored for this stock during WKNephrops in November 2019 ([ICES, 2020](#)). However, MSY reference points could not be derived adequately. Therefore, Nephrops FU30 remains classified as a Category 3 stock.

The SCA method was revisited during WGBIE 2023, following an update of the UWTV survey area and the geostatistical estimation of Nephrops burrow abundance in 2022. In addition, LBI, LBSPR and MLZ methods were also applied during that working group for this stock. However, results from LBI, LBSPR and MLZ should be interpreted with caution, as not all assumptions underlying these methods are met for Nephrops stocks. Key limitations include the assumption of equilibrium conditions, such as constant total mortality and recruitment, which have also been identified in other Nephrops stocks ([Cousido-Rocha et al., 2022](#)). Furthermore, life-history parameters (M/k , L_{inf} , $L50$, $L95$) are highly uncertain, as they have not been updated since the 1980s–1990s and are not specific to FU30. An additional limitation is that these approaches do not explicitly account for spatial structure, which is particularly relevant for Nephrops due to its burrow-dwelling behaviour and strong association with sediment type. Finally, length-composition data should be representative of the exploited population, but recent sampling has been inadequate, with reduced sampling intensity and incomplete quarterly coverage. Some of these considerations also apply when interpreting SCA results.

According to the 2023 ICES guidelines for providing advice on data-limited stocks ([ICES, 2023b](#)), this stock should have been assessed using the ICES **rfb rule** (Method 2.1; ICES ([2021a](#))). However, given the limitations of these methods for this stock and the fishing pressure indicator (HR) accepted during the last benchmark ([ICES, 2017](#)), which provided more complete information than the indicator derived from the new rfb rule, the latter was not applied, following ACOM recommendations.

The Stochastic Surplus Production Model in Continuous Time (SPiCT) has been proposed by ICES to produce MSY advice for several Category 3 stocks in benchmark workshops conducted since 2021. The SPiCT model has already been accepted as the assessment basis for three additional Nephrops functional units in Division 9.a ([González Herraiz et al., 2023](#); [ICES, 2021b](#)), and for one additional Nephrops FU nominated for the next WKMSYSPiCT in 2023–2024.

In addition, to revisiting the advantages and limitations of the scenarios previously tested in past working group meetings, this report extends the exploratory framework by incorporating new SPiCT scenarios that emerged from discussions held after the last meeting in December 2025. Specifically, a new set of scenarios is introduced, all configured consistently with the previous analyses by applying the same four alternative prior configurations (RUN1–RUN4) to each scenario. These new explorations include: (i) an alternative configuration incorporating all available indices simultaneously, (ii) a scenario based on a standardized index formulation, (iii) a scenario using a normalized ARSA survey time series, and (iv) a scenario in which fishing effort is included as a normalized series. The document provides a structured synthesis of these new scenarios and presents their main results, highlighting their relative performance, strengths and limitations in comparison with previously explored configurations, with the aim of informing future benchmark discussions and potential model selection for Nephrops FU30.

Methods

Study Area

The Gulf of Cádiz (FU30, Division 9.a) is located in the southwestern part of the Iberian Peninsula, bordered to the north by the Portuguese coast and to the south by the Strait of Gibraltar (Figure 1). The area is characterised by a wide continental shelf that extends from the coast to depths of approximately 200 m, followed by a steep slope descending to depths exceeding 1 000 m. The seabed is predominantly composed of sandy and muddy sediments, which provide suitable habitats for *Nephrops norvegicus*.

SPiCT description

Surplus productions models are widely used in fisheries stock assessment, particularly for data-limited stocks. These models describe the dynamics of the total biomass of a fish stock over time, considering the balance between growth (production) and removals (catches). The basic equation governing surplus production models is:

$$\frac{dB(t)}{dt} = P(B(t)) - C(t)$$

where: - $B(t)$ is the biomass at time t , - $P(B(t))$ is the surplus production function, representing the net growth of the biomass, - $C(t)$ is the catch at time t .

In a graphical inspection, a surplus production curve typically has a dome-shaped form, indicating that production increases with biomass up to a certain point (the maximum sustainable yield, MSY), after which it declines due to density-dependent effects. The shape of curve depends on the specific functional form chosen for $P(B)$, such as the Schaefer model, Pella-Tomlinson model, or others. (Figure 5).

The SPiCT model extends traditional surplus production models by incorporating stochastic elements and continuous-time dynamics. This allows for a more realistic representation of the uncertainties inherent in fish stock assessments, such as environmental variability and observation errors. The key features of the SPiCT model include: - **Stochasticity**: The model incorporates process and observation errors, allowing for the quantification of uncertainty in biomass estimates and reference points. - **Continuous-time framework**: Unlike discrete-time models, SPiCT operates in continuous time, providing a more flexible and accurate representation of stock dynamics. - **Multiple data sources**: SPiCT can integrate various types of data, including catch data, abundance indices from surveys, and fishery-dependent indices such as LPUE and fishing effort. - **Bayesian inference**: The

model employs Bayesian methods for parameter estimation, enabling the incorporation of prior knowledge and the derivation of posterior distributions for model parameters and stock status indicators.

Data Sources

Fishery description

Nephrops in FU30 is mainly exploited by a single Spanish bottom otter trawl métier (OTB_MCD $\geq$ 55_0_0) and, to a lesser extent, by the Portuguese fleet, operating at depths ranging approximately between 200 and 700 m. The fishery is considered multispecific, targeting a variety of crustaceans, cephalopods and demersal fish, including rose shrimp (*Parapenaeus longirostris*), Nephrops, tiger shrimp, spottail shrimp, octopus, squids, cuttlefish, hake, mullets, sparids, wedge sole, sole and horse mackerel, using a minimum mesh size of 55 mm. Discards are considered negligible.

The increasing abundance of other commercially valuable species in this fishery, particularly rose shrimp (*Parapenaeus longirostris*), is believed to potentially influence fishing behaviour and objectives. Rose shrimp reaches higher market prices and is distributed at shallower depths (approximately 90–380 m) and closer to the coast, making its fishing grounds more accessible.

Landings increased from 1994 to 2003, exceeding 300 t, followed by a sharp decline to 147 t in 2004, representing a reduction of more than 50%. After a temporary recovery in 2005 (246 t), landings declined again to around 120 t in 2008 and stabilised at approximately 100 t until 2012.

During the period 2013–2015, landings decreased dramatically as a result of a penalty imposed by the European Commission for exceeding the TAC in 2012 ([European Union, 2023](#)). The Nephrops fishery was closed for most of 2013, and a substantially reduced TAC (25 t per year) was applied during the subsequent three years. In 2016 and 2017, landings increased to 124 and 140 t, respectively, representing nearly a six-fold increase compared to the penalty period. However, in 2018, landings declined by approximately 46% relative to the previous year, marking the onset of a sustained decreasing trend. This decline continued, with landings in 2023 estimated at 32.8 t.

The progressive reduction of the TAC since 2020 may be constraining the fishery, as evidenced by the closure of the Nephrops fishery from 18 September to 4 December 2023 when landings approached the allocated quota, and the subsequent closure following the exhaustion of the 2023 quota, as established by Regulation (EU) 2024/225 ([European Union, 2024](#)).

Nephrops fishing effort, estimated as the number of trips (fishing days) landing at least 10% Nephrops, shows an increasing trend from 1994 to 2005, when it reached the maximum value of the time series (4 336 fishing days). From 2006 onwards, fishing effort gradually declined, stabilising at approximately 1 500 fishing days until 2012. This reduction in effort was mainly driven by the implementation of successive fishing plans for the Gulf of Cádiz by the Spanish Administration since 2004 (Orders APA/3423/2004, APA/2858/2005, APA/2883/2006, APA/2801/2007, ARM/2515/2009, ARM/58/2010, ARM/2457/2010, AAA/627/2013, AAA/1710/2014, AAA/1406/2016, APM/664/2017 and APM/453/2018).

As a consequence of the sanction imposed in 2012, fishing effort dropped sharply during the period 2013–2015, reaching a mean value of approximately 283 fishing days. Subsequently, fishing effort increased from 2016 (443 fishing days) to 2019 (675 fishing days), remaining relatively stable at around 600 fishing days in 2020 and 2021. However, in 2022 and 2023, Nephrops-directed effort declined again, reaching a mean value of approximately 366 fishing days.

The commercial Nephrops-directed LPUE shows a decreasing trend from 1994 to 2000, followed by fluctuating values for the remainder of the time series. The period 2013–2015 should be interpreted with caution due to the quota overrun penalty imposed in 2012, which increases uncertainty in the LPUE index. In addition, the vessel-level allocation of Nephrops quotas implemented in 2014 may have led to unreported landings, further contributing to uncertainty in the commercial LPUE index. Moreover, since 2016 the commercial LPUE has been estimated using officially reported landings rather than total landings estimated by the Working Group, potentially increasing uncertainty in the index.

Based on the available data analysis and compilation the proposed models for depth-based fisheries correspond to Tier 3, i.e., models of logistic growth for global population productivity, as described by Payá et al. (2014).

Accordingly, two alternatives are presented to model the population dynamics of *Nephrops norvegicus* in the Gulf of Cádiz, as well as to provide recommendations for resource management.

Annual Nephrops landings in FU30 exhibit marked interannual variability over the analysed period (Figure 2). Catches increased steadily during the late 1990s and early 2000s, reaching peak values in the early 2000s, followed by a sharp decline associated with management measures and quota restrictions. After this period, landings remained at comparatively lower levels, with pronounced fluctuations in recent years. The dashed horizontal line in Figure (fig:landings) represents the long-term mean catch and provides a reference against which

periods of above- and below-average exploitation can be identified. Overall, the landings time series reflects both changes in stock availability and the effects of regulatory constraints applied to the fishery.

Abundances Index

The **ISUNEPCA UWTV survey** (U9111) has been conducted annually in the Gulf of Cádiz (FU30) during spring–summer since 2014, although the first survey is considered exploratory. The survey was not carried out in 2020 due to the COVID-19 pandemic. The survey area used to estimate Nephrops abundance in FU30 was originally defined during the Benchmark Workshop on Nephrops stocks (WKNEP) in 2016 (ICES, 2017; Vila, Burgos, & Soriano, 2016) and subsequently modified during WGBIE 2022 (ICES, 2022; Vila & Burgos, 2022). The current area over which Nephrops is distributed covers 2 332.13 km².

The survey follows a randomized isometric grid design, with stations spaced at 4 nautical miles from 2015 to 2021 and reduced to 3.5 nautical miles since 2022. Quantification of Nephrops burrows and the geostatistical estimation of abundance are conducted following ICES Cooperative Research standards (Leocádio, Weetman, & Wieland, 2018).

The methodology described in ICES Cooperative Research Report No. 340 (Leocádio et al., 2018) was used to derive biomass estimates from the UWTV survey. Annual Nephrops biomass was obtained by multiplying the yearly abundance estimates by the annual mean individual weight derived from commercial landings. ICES considers this approach the most appropriate method to obtain absolute biomass estimates for Nephrops stocks and recommends the preferential use of UWTV surveys as the basis for scientific advice for Nephrops (ICES, 2009).

Two bottom trawl **ARSA surveys**, the spring survey (SpSGFS-cspr-WIBTS-Q1) and the autumn survey (SpGFS-caut-WIBTS-Q4), are conducted annually in the southern part of ICES Division 9.a (Gulf of Cádiz), corresponding to FU30. The survey area covers approximately 7 224 km² and spans depths from 15 to 800 m. The sampling design follows a random stratified scheme with five depth strata: 15–30 m, 31–100 m, 101–200 m, 201–500 m and 501–800 m.

These surveys collect information on the distribution, relative abundance and biological characteristics of commercial demersal species, although they are not specifically designed to estimate Nephrops abundance. Nevertheless, they can be used to analyse temporal trends. The Nephrops survey index is expressed as biomass in the two deepest strata (200–500

m and 501–800 m), as *Nephrops* is mainly distributed within these depth ranges, which approximately overlap with the ISUNEPCA UWTV survey area. For some years, data for one or both of these strata are unavailable, leading to the exclusion of the biomass index for those years from the analysis.

The ARSA spring survey (SpSGFS-cspr-WIBTS-Q1) is usually conducted from late February to early March, while the ARSA autumn survey (SpGFS-caut-WIBTS-Q4) takes place in November (Figure ??).

LPUE Standardized The **commercial LPUE series** for *Nephrops* in FU30 has been calculated using landings and effort data from the Spanish and Portuguese fleets operating in the area. Effort is expressed as the number of fishing days, defined as trips landing at least 10% *Nephrops*. The LPUE is calculated as the ratio of landings (in kg) to fishing effort (in fishing days) for each year using as factors.... (...working progress).

A suite of fishery-independent and fishery-dependent indices was used to characterize the temporal dynamics of the *Nephrops* FU30 stock (Figure 3). These include survey-based indices derived from ARSA trawl surveys and ISUNEPCA UWTV surveys, as well as commercial indices such as LPUE and effort-related metrics. The indices display contrasting temporal patterns, reflecting differences in survey design, spatial coverage, and the biological processes captured by each data source. While some indices suggest periods of relative stability, others indicate substantial variability over time, particularly in recent years. Smoothed trends highlight medium-term changes and help to distinguish persistent signals from short-term fluctuations. Taken together, these indices provide complementary information on stock abundance and exploitation dynamics and form the empirical basis for the SPiCT model explorations presented in this study.

Pairwise relationships among the different abundance, productivity and fishery-dependent indices were explored using a Spearman rank correlation analysis. Spearman correlations were selected to account for potential non-linear relationships and non-normal distributions in the time series.

The summary of the input data series used for SPiCT explorations for *Nephrops norvegicus* in Functional Unit 30 is presented in Table 1. The data sources include:

- **Total catches** (1987–2025).
- **Fishing effort** (1994–2025), expressed as total fishing days.
- **Standardised landings per unit of effort (LPUE)** (1994–2025).

- **ISUNEPCA UWTW survey indices**, including:
 - absolute **abundance index** (available from 2015–2025), and
 - associated **biomass index** (available for selected years).
- **ARSA survey indices**, comprising:
 - **spring survey index** (1993–2025),
 - and derived **productivity indices** used in selected scenarios.

All time series were harmonised temporally and formatted consistently prior to model fitting. Alternative combinations of these data sources were used to define the different SPiCT scenarios explored in this study.

Table 1: Summary of input data series used for SPiCT explorations for *Nephrops norvegicus* in Functional Unit 30. Column abbreviations are as follows: CATC = total catches (t); ARSB = ARSA biomass survey index; ARSP = ARSA productivity index; ARSS = standardised ARSA index; ARCV = coefficient of variation of the ARSA index; ISUB = ISUNEPCA UWTV biomass index; ISUA = ISUNEPCA UWTV abundance index; EFRT = fishing effort (days); and LPUE = standardised landings per unit of effort.

Year	CATC	ARSB	ARSP	ARSS	ARCV	ISUB	ISUA	EFRT	LPUE
1987	193.0000	NA	NA	NA	NA	NA	NA	NA	NA
1988	138.6700	NA	NA	NA	NA	NA	NA	NA	NA
1989	174.5000	NA	NA	NA	NA	NA	NA	NA	NA
1990	220.1000	NA	NA	NA	NA	NA	NA	NA	NA
1991	225.4010	NA	NA	NA	NA	NA	NA	NA	NA
1992	242.7600	NA	NA	NA	NA	NA	NA	NA	NA
1993	159.7000	26.603058	0.95	0.7798737	0.7944869	NA	NA	NA	NA
1994	107.6180	28.159856	0.76	NA	NA	NA	NA	27823	0.0038680
1995	130.6230	NA	NA	NA	NA	NA	NA	32194	0.0040574
1996	48.5220	24.615855	0.93	0.4628532	0.9384557	NA	NA	31951	0.0015186
1997	97.0760	9.328451	0.38	0.2898361	1.1413823	NA	NA	32573	0.0029803
1998	85.3150	9.301283	0.30	0.1514835	1.0490586	NA	NA	32824	0.0025992
1999	120.2170	NA	0.41	0.4627855	1.0890912	NA	NA	32731	0.0036729
2000	128.8620	10.140252	0.37	0.4303017	0.8514907	NA	NA	29875	0.0043134
2001	178.4260	12.331178	0.44	0.4807538	0.8150957	NA	NA	30416	0.0058662
2002	261.8360	10.871560	0.47	0.6709178	0.9846021	NA	NA	31760	0.0082442
2003	306.5170	NA	NA	NA	NA	NA	NA	32790	0.0093479
2004	147.0209	4.240238	0.15	0.3125780	1.1381654	NA	NA	30901	0.0047578
2005	246.1790	17.429929	0.64	0.7437840	0.9839900	NA	NA	33203	0.0074144
2006	245.3679	12.350958	0.42	0.5396657	0.8885340	NA	NA	26946	0.0091059
2007	214.4429	10.541994	0.37	0.4733010	0.8630268	NA	NA	25366	0.0084540

Table 1: Summary of input data series used for SPiCT explorations for *Nephrops norvegicus* in Functional Unit 30. Column abbreviations are as follows: CATC = total catches (t); ARSB = ARSA biomass survey inde (*continued*)

Year	CATC	ARSB	ARSP	ARSS	ARCV	ISUB	ISUA	EFRT	LPUE
2008	119.7816	23.930662	0.85	0.8523799	0.8825238	NA	NA	20602	0.0058141
2009	119.2902	13.137462	0.37	0.3372162	1.0435310	NA	NA	21401	0.0055740
2010	107.3343	NA	NA	1.0786173	1.1511345	NA	NA	21352	0.0050269
2011	95.9099	6.974094	0.23	0.4699803	1.1842075	NA	NA	19986	0.0047989
2012	116.0000	5.038396	0.18	0.2564917	1.3004164	NA	NA	20402	0.0056857
2013	26.7170	23.155208	0.79	0.3648117	1.0193968	NA	NA	23099	0.0011566
2014	14.8850	48.654413	1.92	1.2084349	1.0159162	NA	NA	NA	NA
2015	24.6410	33.141152	1.26	1.7672853	0.9556471	NA	NA	NA	NA
2016	123.7210	68.015128	2.73	3.1413183	0.9831106	1.1162106	100.340	22458	0.0055090
2017	139.5160	40.933017	1.67	1.4951162	1.1421458	0.8661148	77.270	21583	0.0064642
2018	75.5510	9.263278	0.38	0.4755977	0.7913575	1.1290788	100.180	20748	0.0036414
2019	64.7700	32.037073	1.34	2.0268816	1.0653751	1.7104377	109.933	20553	0.0031514
2020	62.8690	165.458399	6.16	5.4609633	0.8724866	NA	NA	19991	0.0031449
2021	49.2760	NA	NA	NA	NA	1.2472901	66.238	20443	0.0024104
2022	44.4150	6.716924	0.25	0.2596278	1.3394774	0.7805596	53.110	18853	0.0023559
2023	32.8630	17.925767	0.67	0.6070566	0.8795903	0.7633243	54.193	19671	0.0016706
2024	32.0470	102.613914	4.08	2.4000868	0.8363039	0.8525884	63.698	19240	0.0016656
2025	NA	NA	NA	NA	NA	0.5343957	38.023	NA	NA

Previous Nephrops FU30 Scenarios

Several SPiCT model configurations were explored for Nephrops FU30, varying in the inclusion of abundance indices, catch data, and model parameterisations (priors). This exploratory framework is consistent with the approach adopted in previous ICES Nephrops working groups, where alternative data configurations and prior assumptions have been systematically tested to evaluate model robustness and suitability for assessment purposes. The full set of SPiCT model configurations explored in this study is summarised in Table 2, including the input data used in each scenario and the corresponding prior configurations applied consistently across scenarios. This structured exploration allows for a transparent comparison of model behaviour and performance, in line with recent methodological developments discussed within ICES Nephrops expert groups.

The **Scenario 0** included catch data from 1994 to 2023 and the ISUNEPCA UWTV survey index from 2015 to 2023, reflecting the core data sources currently used in the assessment framework for this stock. Additional scenarios were developed to assess the effect of incorporating supplementary fishery-independent and fishery-dependent information, such as the ARSA survey index (**Scenario 2**), as well as alternative indicators derived from effort and LPUE series. Further scenarios explored the sensitivity of model outcomes to different assumptions regarding process error, observation error, and prior distributions for key biological parameters, following configurations commonly examined in recent Nephrops assessments (**Scenarios 3 and 4**).

Explorations were conducted using data available up to 2025. Tables below summarise the different scenarios and model configurations tested. Fishing effort and LPUE were not included in any scenario, as survey-based indices are recommended when available. The autumn ARSA survey (SpGFS-caut-WIBTS-Q4) was also excluded from the analyses, as part of the stock is not accessible to the gear during this survey due to the reproductive behaviour of Nephrops. In November, ovigerous females remain inside their burrows and are therefore unavailable to bottom trawl sampling. For this reason, the autumn ARSA survey index has never been included in the FU30 WGBIE assessments.

The ISUNEPCA UWTV survey index is considered the most reliable abundance index for Nephrops in FU30 and is currently used as the basis for assessment and advice (ICES, 2023a). This survey is conducted in the year of the assessment; for example, the UWTV survey planned for June 2024 will be used to inform the advice for 2025, delivered in October 2024. If the 2024 ISUNEPCA index were to be included in a SPiCT model, catch data for 2024 would not yet be available. This issue should be explicitly addressed if SPiCT is adopted as

the assessment model for this stock in a future benchmark.

Table 2: Summary of SPiCT model configurations (scenarios) explored for *Nephrops norvegicus* in Functional Unit 30. All scenarios were tested under four alternative prior configurations (RUN1–RUN4).

Scenario	Input data			Time span	Methodological description
SC0	Catches;	ISUNEPCA	UWTV	1987–2025	Baseline configuration using only the UWTV survey as absolute abundance index.
SC1	Catches;	ISUNEPCA	UWTV	1987–2025	Configuration combining UWTV and ARSA surveys to assess consistency between fishery-independent indices.
SC2	Catches;	ISUNEPCA	UWTV	1987–2025	Configuration including fishery-dependent information through directed LPUE.
SC3	Catches;	ISUNEPCA	UWTV	1987–2025	Configuration using directed fishing effort as a proxy of stock dynamics.
SC4	Catches;	ISUNEPCA	UWTV	1987–2025	Configuration testing the use of standardized LPUE instead of raw commercial indices.

New Scenarios in Nephrops FU30

To extend the SPiCT-based assessment of *Nephrops*, we defined **five additional modelling scenarios (Scenarios 5–9)** that incorporate alternative combinations of fishery-dependent and fishery-independent information. *Scenario 5* combines total landings with the **normalized ISUNEPCA UWTV biomass index** and the **normalized ARSA biomass index**, providing a configuration based exclusively on survey-derived biomass information. *Scenario 6* integrates total landings with the **normalized ISUNEPCA UWTV biomass index** and a **standardised LPUE**, allowing direct comparison between fishery-independent biomass and fishery-dependent abundance signals. *Scenario 7* represents a more information-rich configuration that combines **normalized ISUNEPCA biomass**, **normalized ARSA biomass**, and **standardised LPUE** together with landings. *Scenario 8* combines landings with the **normalized ISUNEPCA UWTV abundance index** and **standardised LPUE**, providing an alternative representation of stock dynamics based on abundance rather than biomass. The definition of *Scenario 9* is motivated by a major structural change in the ARSA survey. In 2012, the ARSA survey programme experienced a change of vessel and operational characteristics, introducing a potential discontinuity in the biomass index methodology estimation. To avoid confounding biological signals with survey artefacts, only the pre-2012 ARSA biomass series is retained in this scenario. Post-2012 stock dynamics are instead informed by the ISUNEPCA UWTV survey, which provides a consistent biomass index from 2015 onward, and by the standardised LPUE, which reflects commercial fleet performance in recent years. This scenario therefore provides a coherent data configuration that respects survey continuity while preserving both long-term and contemporary information for stock assessment (Table 3). All scenarios include total landings as the catch time series and differ in the configuration of abundance indices (Figure 3).

Table 3: Extended SPiCT model configurations (Scenarios 6–8) explored for *Nephrops norvegicus* in Functional Unit 30. All scenarios were fitted using the same four alternative prior configurations (RUN1–RUN4).

Scenario	Input data	Time span	Methodological description
SC5	Catches; ISUNEPCA UWTV biomass (normalized); ARSA biomass (normalized)	1987–2025	Scenario combining total landings with normalized ISUNEPCA UWTV biomass and normalized ARSA biomass indices, representing a configuration based exclusively on fishery-independent information.
SC6	Catches; ISUNEPCA UWTV biomass (normalized); Standardised LPUE	1987–2025	Scenario integrating total landings with normalized ISUNEPCA UWTV biomass and a standardized LPUE index, combining fishery-independent and fishery-dependent information.

Table 3: Extended SPiCT model configurations (Scenarios 6–8) explored for *Nephrops norvegicus* in Functional Unit 30. All scenarios were fitted using the same four alternative prior configurations (RUN1–RUN4). (*continued*)

Scenario	Input data	Time span	Methodological description
SC7	Catches; ISUNEPCA UWTB biomass (normalized); ARSA biomass (normalized); Standardised LPUE	1987–2025	Most information-rich configuration, combining total landings with normalized ISUNEPCA UWTB biomass, normalized ARSA biomass, and standardized LPUE indices to explore model behaviour under maximum data availability.
SC8	Catches; ISUNEPCA UWTB abundance (normalized); Standardised LPUE	1987–2025	Scenario integrating total landings with normalized ISUNEPCA UWTB abundance and a standardized LPUE index, combining fishery-independent and fishery-dependent information.
SC9	CATCHES; ISUNEPCA UWTB biomass (normalized), ARSA biomass (normalized), Standardised LPUE	1987–2025	Scenario combines total landings with the ISUNEPCA UWTB biomass index (2015–2025), ARSA biomass survey (1993–2012), and the standardised commercial LPUE (2009–2024). This configuration integrates long-term and recent information while ensuring consistency among data sources.

Across all SPiCT scenarios, four alternative prior configurations (RUN1–RUN4) were explored to evaluate model sensitivity to assumptions on stock depletion, production curve shape, and intrinsic population growth. The baseline run (RUN1) relied on default SPiCT priors, while subsequent configurations progressively introduced informative priors on initial depletion (B/K), the production curve shape parameter (n), and the intrinsic growth rate (r). This hierarchical design ensured consistent and comparable evaluation of prior influence across scenarios (Table 4).

Table 4: Summary of prior configurations (RUNs) explored for the SPiCT model applied to *Nephrops norvegicus* in Functional Unit 30.

Run	Prior configuration	Specification	Purpose
RUN1	Default (package defaults)	Default SPiCT priors	Baseline configuration using default SPiCT assumptions.
RUN2	logbkfrac	$\logbkfrac \sim \log(0.5)$, SD = 0.2	Constrains initial depletion (B/K) around an intermediate level with low uncertainty.
RUN3	logbkfrac, logn	$\logbkfrac \sim \log(0.5)$, SD = 0.2; $\logn \sim \log(2)$, SD = 0.5	Adds a tighter prior on the production curve shape parameter (n).
RUN4	logbkfrac, logn, logr	$\logbkfrac \sim \log(0.5)$, SD = 0.2; $\logn \sim \log(2)$, SD = 0.5; $\logr \sim \log(0.2)$, SD = 0.2	Fully informative configuration including priors on depletion, production curve shape and intrinsic growth rate.

To ensure consistency and comparability across scenarios, all new configurations are fitted using the same set of four prior specifications previously defined in the analysis. Results will

be presented comparatively across scenarios, including diagnostic plots, key stock status outputs, and management-related figures and summary tables, allowing a coherent evaluation of model behaviour and management implications under alternative data combinations. A total of **36 SPiCT model runs** were conducted, corresponding to four alternative prior configurations (RUN1–RUN4) applied consistently across nine model scenarios (SC0–SC8). This factorial design allowed for the evaluation of model behaviour, convergence and sensitivity to both data configuration and prior assumptions on key biological and production parameters.

Results

Previous Analysis

The exploratory modelling scenarios SC0 to SC5 implemented in the present analysis are documented in detail in the technical report by Vila & Quinzán (2024), where alternative configurations of fishery-dependent and fishery-independent data were systematically evaluated using the SPiCT framework for *Nephrops norvegicus* in Functional Unit 30. These scenarios form part of the broader exploratory analyses conducted to assess model behaviour, data sensitivity, and the robustness of stock status estimates under different information assumptions.

In the new explorations, in this report we focus on the results obtained from the additional scenarios (SC6 to SC8) that incorporate alternative combinations of abundance indices and fishery-dependent information. The following sections present a synthesis of the key findings from these new scenarios, including diagnostic evaluations, stock dynamics, and management implications, while also referencing relevant results from the previously documented scenarios (SC0 to SC5) for comparative purposes.

New Scenarios (5-9)

To optimise the presentation and interpretation of model outputs for the SPiCT assessment of *Nephrops norvegicus* in Functional Unit 30, only scenarios and prior configurations that successfully converged are retained for reporting. For these converged runs, we present the key summary tables and diagnostic and management relevant figures, ensuring that all results shown are based on numerically stable and internally consistent model solutions. This approach facilitates a clear and robust comparison of alternative data configurations and prior assumptions while avoiding misleading inferences from non-convergent model fits.

Correlation analysis of input data series The Figure 4 reveals strong positive correlations among indices derived from similar sources, particularly between ISUNEPCA UWTV biomass and abundance indices, as well as between standardized LPUE and effort-related metrics. In contrast, ARSA-based indices show weaker or negative correlations with some fishery-dependent indicators, highlighting differences in spatial coverage and survey design. Overall, the results indicate a clear clustering of indices by data origin, suggesting partial redundancy among some series and complementarity across others.

Hindcast Analysis

The Mean Absolute Scaled Error (MASE) is a widely used metric for evaluating the predictive accuracy of time series models. It is defined as the mean absolute error of the model's predictions, scaled by the mean absolute error of a naive forecast (typically the previous observation). MASE values below 1 indicate that the model outperforms the naive forecast, while values above 1 suggest inferior performance.

Hindcast diagnostics by scenario and prior configuration are presented in Figures 7 and 8–15, illustrating model predictive performance under alternative data and prior combinations.

Retrospective Analysis

Retrospective patterns for each modelling scenario were evaluated under alternative prior configurations. Scenario-specific diagnostics are shown in Figures 16–33, illustrating the temporal stability of biomass and fishing mortality estimates across successive peelings. Retrospective bias was evaluated using Mohn's rho estimates (Table 5).

The results indicate that all model runs achieved MASE values below 1, demonstrating that the SPiCT model consistently outperformed the naive forecast in predicting stock dynamics. Notably, RUN3 exhibited the lowest MASE value, suggesting that the inclusion of informative priors on both initial depletion and production curve shape enhanced model predictive accuracy. These findings underscore the importance of prior specification in improving model performance and provide confidence in the SPiCT model's ability to capture key stock dynamics for Nephrops FU30.

Comparative Population variables

Scenario-specific stock dynamics derived from the SPiCT model are illustrated through estimated biomass (Figures ??–??) and fishing mortality trajectories (Figures ??–??), allowing direct comparison of model behaviour across alternative data configurations.

Biological Reference Points.

The Figure 6 compares key biological reference points across **Scenarios SC5–SC9** and **prior configurations (RUN1–RUN4)** for Nephrops FU30. Several clear patterns emerge. Estimates of **Bmsy** and **MSY** vary widely across RUNs within the same scenario. RUN1–RUN3 typically yield **much higher Bmsy and MSY** than RUN4, indicating that the more informative priors in RUN4 (on depletion, productivity, and growth) strongly constrain stock productivity and carrying capacity. This implies that data alone are not sufficient to tightly

identify absolute scale, and priors play a major role in determining long-term productivity. Across scenarios, **Fmsy** is far less variable than **Bmsy** or **MSY**, particularly under RUN4, where Fmsy consistently clusters around **~0.12–0.15**. This suggests that exploitation rate at MSY is relatively well informed by the time series, even when absolute biomass is uncertain.

Scenarios **SC7 and SC8**, which combine **normalized UWTV biomass, ARSA biomass, and/or standardized LPUE**, show tighter clustering across RUNs compared with SC5–SC6. Adding complementary fishery-independent and fishery-dependent indices improves identifiability, especially for Fmsy and Cp. SC9 displays extreme or unstable MSY and Bmsy for some RUNs, reflecting the mixed time coverage (UWTV 2015–2025, ARSA biomass 1993–2012, LPUE 2009–2024) and the survey break in 2012. This configuration increases structural uncertainty and makes results more prior-dependent.

Catch suggested is broadly similar across RUNs within scenarios, indicating that recent catches are well constrained by data and largely independent of prior assumptions.

(Table with main outputs, Bmsy, Fmsy and MSY)

Management Outputs

Management strategy evaluation results by scenario are summarised in Table 7. Values correspond to equilibrium projections for relative biomass (B/B_{msy}), fishing mortality (F/F_{msy}), and expected catches under each management rule.

CHECK!!! Key reference points, including MSY, BMSY, FMSY, and current stock status indicators (B/B_{MSY} and F/F_{MSY}), are presented alongside uncertainty estimates (95% credible intervals). The results indicate that all scenarios suggest the stock is currently below BMSY, with varying degrees of fishing pressure relative to FMSY. Notably, Scenario 2 under RUN4 configuration indicates the most optimistic stock status, with B/B_{MSY} approaching 1 and F/F_{MSY} below 1, suggesting potential for recovery under current management measures. In contrast, Scenario 0 under RUN1 configuration indicates a more depleted stock status, with B/B_{MSY} significantly below 1 and F/F_{MSY} exceeding 1, highlighting the need for continued management interventions. Overall, the results underscore the sensitivity of stock status estimates to both data configuration and prior assumptions, emphasizing the importance of robust model selection in fisheries management.

Conclusions

WGBIE considered the results obtained for Nephrops FU 30 stock promising and will propose it as possible candidate for the next WKMSYSPiCT benchmark. However, further work should be conducted to improve the model according to the most recent SPiCT guidelines ([Mildenberger, Kokkalis, & Berg, 2023](#)). The WKMSYSPiCT benchmark, where a close contact with SPiCT expert is possible, could help to find the best model configuration to provide the assessment for this stock. Further work is needed to improve the model, including:

- Extend the time series of landings to earlier years in order to improve the stability and robustness of the model
- Add uncertainty in some years of the biomass indices time series or catches
- Fine-tune model configuration.

Implication for management and benchmarking RUN4 provides the most conservative and stable estimates of productivity and fishing mortality across scenarios. SC7–SC8 appear to offer the best balance between information content and robustness, whereas SC9 should be treated cautiously due to survey discontinuity–driven uncertainty.

Code Repositories

All this work has been developed using R language. The code and data used for the SPiCT explorations for Nephrops FU30 can be found in the following GitHub repository:

All updates and improvements can be followed and obtained at the [SPiCT GitHub repository](#). Bugs and issues can be reported via the [SPiCT Releases page](#).

Nephrops Stock Assessment and Exploratory Data Analysis (ICES Division 9A) codes to replicate part of this analysis, is available at the following GitHub repository: [Nephros SPiCT codes](#)

Figures and Tables

(Que figura sería interesante incorporar aparte de las q están?)

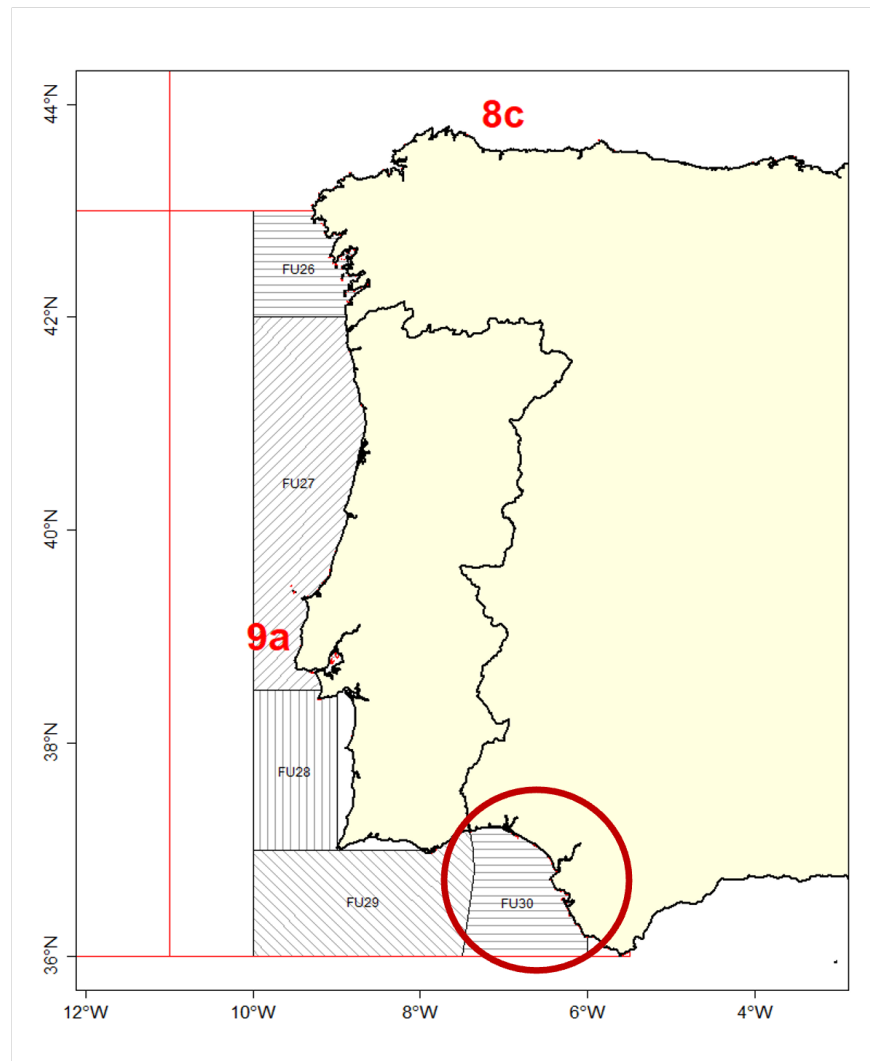


Figure 1: Map of the Gulf of Cádiz (FU30, Division 9.a) showing the main fishing grounds for *Nephrops norvegicus*.

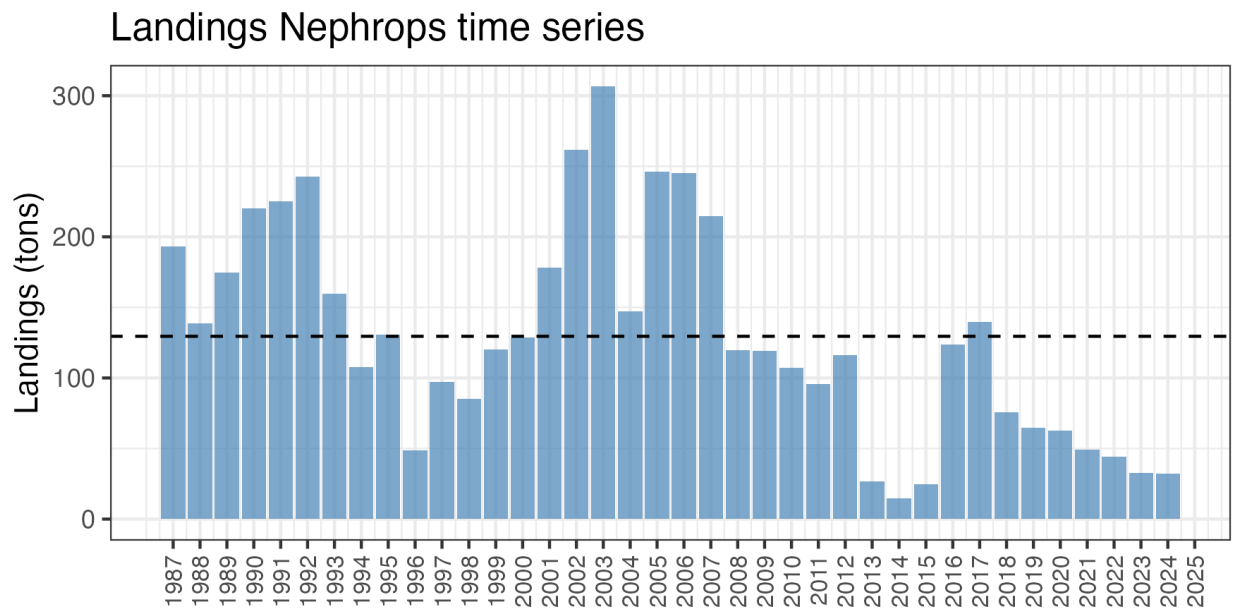


Figure 2: Time series of annual Nephrops landings (tons) for FU30 (Division 9.a). Bars represent reported annual catches, while the dashed horizontal line indicates the long-term mean catch over the period analysed.

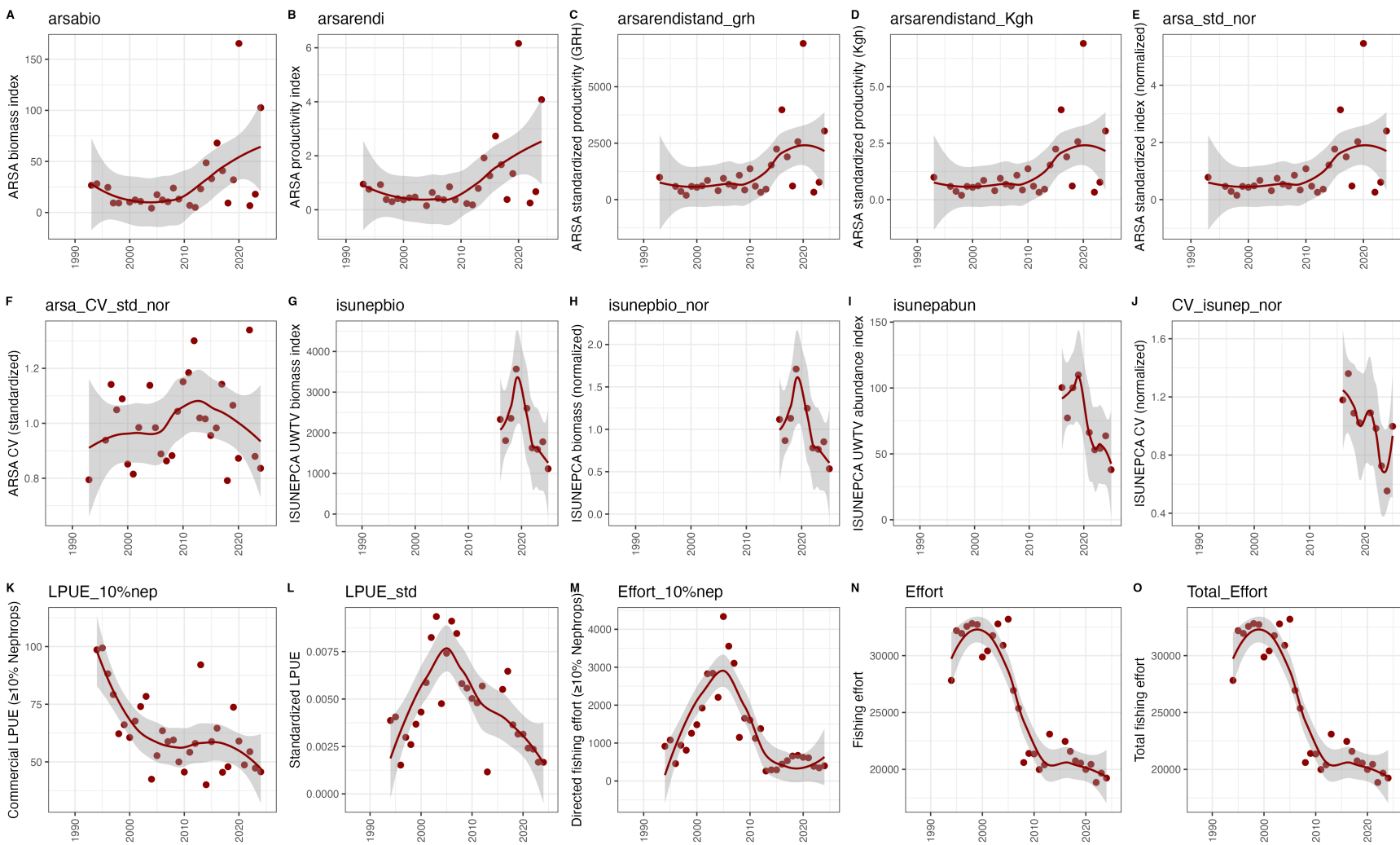


Figure 3: Time series of abundance and fishery-dependent indices used in the SPiCT explorations for Nephrops FU30 (Division 9.a). Panels A–H show the different survey-based and commercial indices. Points represent observed annual values, while solid lines correspond to smoothed trends with associated uncertainty envelopes.

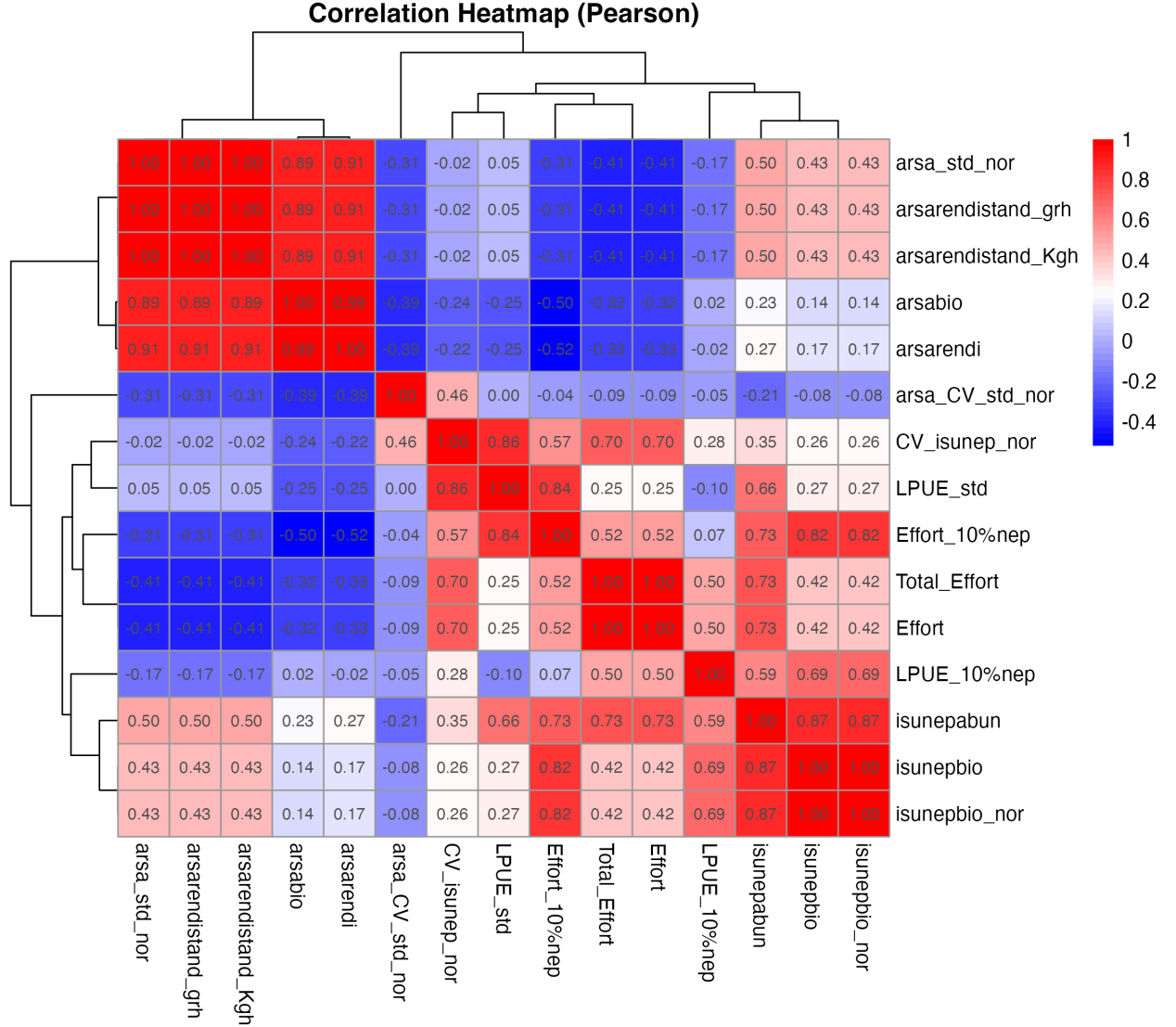


Figure 4: Spearman rank correlation heatmap among abundance, productivity and fishery-dependent indices for *Nephrops* FU30 (Division 9.a). Colours indicate the strength and direction of correlations (red: positive; blue: negative), and hierarchical clustering highlights groups of highly correlated indices.

Table 5: Mohn's rho estimates derived from retrospective analyses of the SPiCT models for *Nephrops norvegicus* in Functional Unit 30. Values are reported for relative spawning biomass (B/B_{MSY}) and fishing mortality (F/F_{MSY}) by scenario and prior configuration (RUN).

Scenario	Run	Mohn_BBmsy	Mohn_FFmsy
SC0	RUN4	-0.22	1.87
SC1	RUN1	-0.28	1.19
SC1	RUN2	-0.08	0.19
SC1	RUN3	-0.07	0.12

Table 5: Mohn's rho estimates derived from retrospective analyses of the SPiCT models for *Nephrops norvegicus* in Functional Unit 30. Values are reported for relative spawning biomass (B/B_{MSY}) and fishing mortality (F/F_{MSY}) by scenario and prior configuration (RUN). (*continued*)

Scenario	Run	Mohn_BBmsy	Mohn_FFmsy
SC1	RUN4	-0.07	0.15
SC2	RUN1	0.02	0.87
SC2	RUN2	NA	NA
SC2	RUN3	NA	NA
SC2	RUN4	0.12	-0.15
SC3	RUN4	0.00	0.00
SC4	RUN1	-0.31	0.12
SC4	RUN2	0.00	-0.09
SC4	RUN3	0.01	-0.10
SC4	RUN4	0.07	-0.27
SC5	RUN1	45.73	NA
SC5	RUN2	0.83	NA
SC5	RUN3	NA	0.01
SC5	RUN4	0.01	-0.07

Table 7: Projected stock status and catch outcomes under alternative management rules by scenario and prior configuration (RUN) for the SPiCT-based *Nephrops* FU30 assessment.

Management_rule	Catch	B/Bmsy	F/Fmsy
Scenario SC0 – RUN1			
1. Keep current catch	33.0	1.45	0.16
2. Keep current F	28.1	1.46	0.14
3. Fish at Fmsy	186.2	1.19	1.00
4. No fishing	0.0	1.51	0.00
5. Reduce F by 25%	21.2	1.47	0.10
6. Increase F by 25%	35.0	1.45	0.17
7. MSY hockey-stick rule	186.2	1.19	1.00
8. ICES advice rule	146.5	1.25	0.77
Scenario SC0 – RUN2			
1. Keep current catch	32.9	1.45	0.14
2. Keep current F	28.9	1.46	0.12
3. Fish at Fmsy	207.9	1.17	1.00
4. No fishing	0.0	1.50	0.00
5. Reduce F by 25%	21.8	1.47	0.09
6. Increase F by 25%	36.0	1.45	0.16
7. MSY hockey-stick rule	207.9	1.17	1.00
8. ICES advice rule	163.1	1.25	0.76
Scenario SC0 – RUN3			
1. Keep current catch	32.9	1.45	0.14
2. Keep current F	28.8	1.45	0.13
3. Fish at Fmsy	206.1	1.18	1.00

Table 7: Projected stock status and catch outcomes under alternative management rules by scenario and prior configuration (RUN) for the SPiCT-based Nephrops FU30 assessment.
(continued)

Management_rule	Catch	B/Bmsy	F/Fmsy
4. No fishing	0.0	1.50	0.00
5. Reduce F by 25%	21.7	1.46	0.09
6. Increase F by 25%	35.8	1.44	0.16
7. MSY hockey-stick rule	206.1	1.18	1.00
8. ICES advice rule	161.6	1.25	0.76
Scenario SC0 – RUN4			
1. Keep current catch	33.1	1.54	0.05
2. Keep current F	24.0	1.54	0.04
3. Fish at Fmsy	586.0	1.40	1.00
4. No fishing	0.0	1.54	0.00
5. Reduce F by 25%	18.0	1.54	0.03
6. Increase F by 25%	30.0	1.54	0.05
7. MSY hockey-stick rule	586.0	1.40	1.00
8. ICES advice rule	460.0	1.43	0.78
Scenario SC1 – RUN1			
1. Keep current catch	32.5	3.23	0.05
2. Keep current F	26.7	3.26	0.04
3. Fish at Fmsy	473.1	1.44	1.00
4. No fishing	0.0	3.37	0.00
5. Reduce F by 25%	20.1	3.29	0.03
6. Increase F by 25%	33.2	3.23	0.05
7. MSY hockey-stick rule	473.1	1.44	1.00
8. ICES advice rule	384.9	1.79	0.73
Scenario SC1 – RUN2			
1. Keep current catch	32.5	3.05	0.05
2. Keep current F	26.2	3.07	0.04
3. Fish at Fmsy	462.0	1.56	1.00
4. No fishing	0.0	3.16	0.00
5. Reduce F by 25%	19.7	3.10	0.03
6. Increase F by 25%	32.6	3.05	0.05
7. MSY hockey-stick rule	462.0	1.56	1.00
8. ICES advice rule	371.7	1.87	0.74
Scenario SC1 – RUN3			
1. Keep current catch	32.6	1.71	0.10
2. Keep current F	26.5	1.72	0.08
3. Fish at Fmsy	280.8	1.38	1.00
4. No fishing	0.0	1.76	0.00
5. Reduce F by 25%	19.9	1.73	0.06
6. Increase F by 25%	33.1	1.71	0.11
7. MSY hockey-stick rule	280.8	1.38	1.00
8. ICES advice rule	216.7	1.46	0.75
Scenario SC1 – RUN4			
1. Keep current catch	32.5	1.62	0.10
2. Keep current F	24.7	1.62	0.08
3. Fish at Fmsy	309.4	1.45	1.00

Table 7: Projected stock status and catch outcomes under alternative management rules by scenario and prior configuration (RUN) for the SPiCT-based Nephrops FU30 assessment.

(continued)

Management_rule	Catch	B/Bmsy	F/Fmsy
4. No fishing	0.0	1.64	0.00
5. Reduce F by 25%	18.6	1.62	0.06
6. Increase F by 25%	30.9	1.62	0.09
7. MSY hockey-stick rule	309.4	1.45	1.00
8. ICES advice rule	235.2	1.50	0.75
Scenario SC2 – RUN1			
1. Keep current catch	32.2	2.00	0.01
2. Keep current F	32.3	2.00	0.01
3. Fish at Fmsy	3103.3	1.82	1.00
4. No fishing	0.0	2.01	0.00
5. Reduce F by 25%	24.2	2.01	0.01
6. Increase F by 25%	40.4	2.00	0.01
7. MSY hockey-stick rule	3103.3	1.82	1.00
8. ICES advice rule	2351.1	1.86	0.75
Scenario SC2 – RUN2			
1	NA	NA	NA
Scenario SC2 – RUN3			
1	NA	NA	NA
Scenario SC2 – RUN4			
1. Keep current catch	32.3	0.51	0.52
2. Keep current F	32.7	0.51	0.53
3. Fish at Fmsy	61.6	0.50	1.00
4. No fishing	0.0	0.52	0.00
5. Reduce F by 25%	24.6	0.51	0.39
6. Increase F by 25%	40.7	0.51	0.66
7. MSY hockey-stick rule	60.3	0.50	0.98
8. ICES advice rule	44.2	0.51	0.71
Scenario SC3 – RUN1			
1	NA	NA	NA
Scenario SC3 – RUN2			
1	NA	NA	NA
Scenario SC3 – RUN3			
1	NA	NA	NA
Scenario SC3 – RUN4			
1. Keep current catch	90.7	0.51	1.00
2. Keep current F	91.1	0.51	1.01
3. Fish at Fmsy	90.9	0.51	1.00
4. No fishing	0.2	0.51	0.00
5. Reduce F by 25%	68.3	0.51	0.75
6. Increase F by 25%	113.8	0.51	1.26
7. MSY hockey-stick rule	90.9	0.51	1.00
8. ICES advice rule	87.1	0.51	0.96
Scenario SC4 – RUN1			
1. Keep current catch	40.4	0.00	12.60

Table 7: Projected stock status and catch outcomes under alternative management rules by scenario and prior configuration (RUN) for the SPiCT-based Nephrops FU30 assessment.

(continued)

Management_rule	Catch	B/Bmsy	F/Fmsy
2. Keep current F	55.2	0.00	4.18
3. Fish at Fmsy	31.9	0.00	1.00
4. No fishing	0.2	0.01	0.00
5. Reduce F by 25%	54.2	0.00	3.13
6. Increase F by 25%	53.9	0.00	5.22
7. MSY hockey-stick rule	0.1	0.01	0.00
8. ICES advice rule	0.0	0.01	0.00
Scenario SC4 – RUN2			
1. Keep current catch	35.5	0.68	0.62
2. Keep current F	41.6	0.66	0.73
3. Fish at Fmsy	55.3	0.62	1.00
4. No fishing	0.0	0.77	0.00
5. Reduce F by 25%	31.8	0.69	0.55
6. Increase F by 25%	51.0	0.64	0.91
7. MSY hockey-stick rule	55.3	0.62	1.00
8. ICES advice rule	44.2	0.65	0.78
Scenario SC4 – RUN3			
1. Keep current catch	35.4	0.66	0.64
2. Keep current F	41.3	0.65	0.76
3. Fish at Fmsy	53.3	0.61	1.00
4. No fishing	0.0	0.75	0.00
5. Reduce F by 25%	31.5	0.67	0.57
6. Increase F by 25%	50.6	0.62	0.95
7. MSY hockey-stick rule	53.3	0.61	1.00
8. ICES advice rule	42.6	0.64	0.78
Scenario SC4 – RUN4			
1. Keep current catch	35.3	0.92	0.36
2. Keep current F	36.4	0.91	0.37
3. Fish at Fmsy	95.7	0.86	1.00
4. No fishing	0.0	0.95	0.00
5. Reduce F by 25%	27.4	0.92	0.28
6. Increase F by 25%	45.3	0.91	0.46
7. MSY hockey-stick rule	95.7	0.86	1.00
8. ICES advice rule	77.0	0.88	0.80
Scenario SC5 – RUN1			
1	NA	NA	NA
Scenario SC5 – RUN2			
1	NA	NA	NA
Scenario SC5 – RUN3			
1. Keep current catch	32.5	2.61	0.06
2. Keep current F	34.3	2.60	0.06
3. Fish at Fmsy	371.0	1.35	1.00
4. No fishing	0.0	2.71	0.00
5. Reduce F by 25%	25.9	2.63	0.05
6. Increase F by 25%	42.6	2.58	0.08

Table 7: Projected stock status and catch outcomes under alternative management rules by scenario and prior configuration (RUN) for the SPiCT-based Nephrops FU30 assessment.

(continued)

Management_rule	Catch	B/Bmsy	F/Fmsy
7. MSY hockey-stick rule	371.0	1.35	1.00
8. ICES advice rule	303.5	1.64	0.73
Scenario SC5 – RUN4			
1. Keep current catch	32.5	0.62	0.41
2. Keep current F	26.0	0.63	0.33
3. Fish at Fmsy	78.0	0.60	1.00
4. No fishing	0.0	0.64	0.00
5. Reduce F by 25%	19.6	0.63	0.24
6. Increase F by 25%	32.4	0.62	0.41
7. MSY hockey-stick rule	78.0	0.60	1.00
8. ICES advice rule	58.8	0.61	0.75

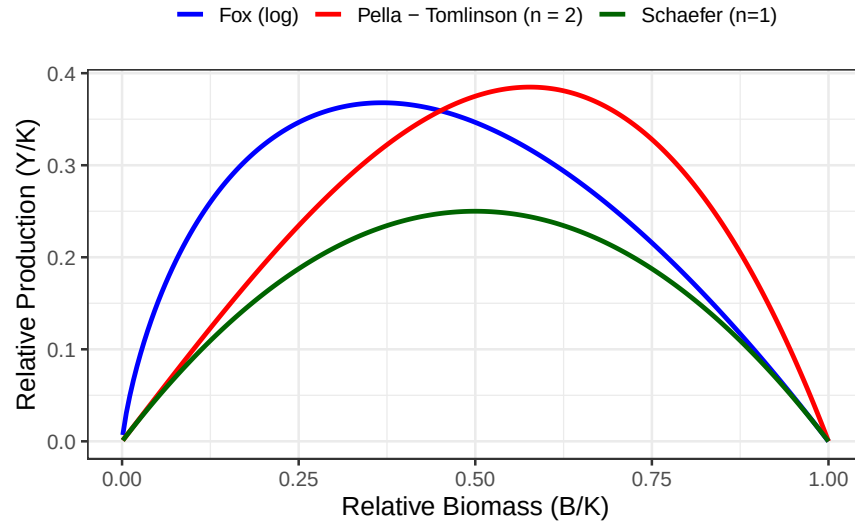


Figure 5: Comparison of surplus production functions: Schaefer (n=1), Pella - Tomlinson (n=2) and Fox (log).

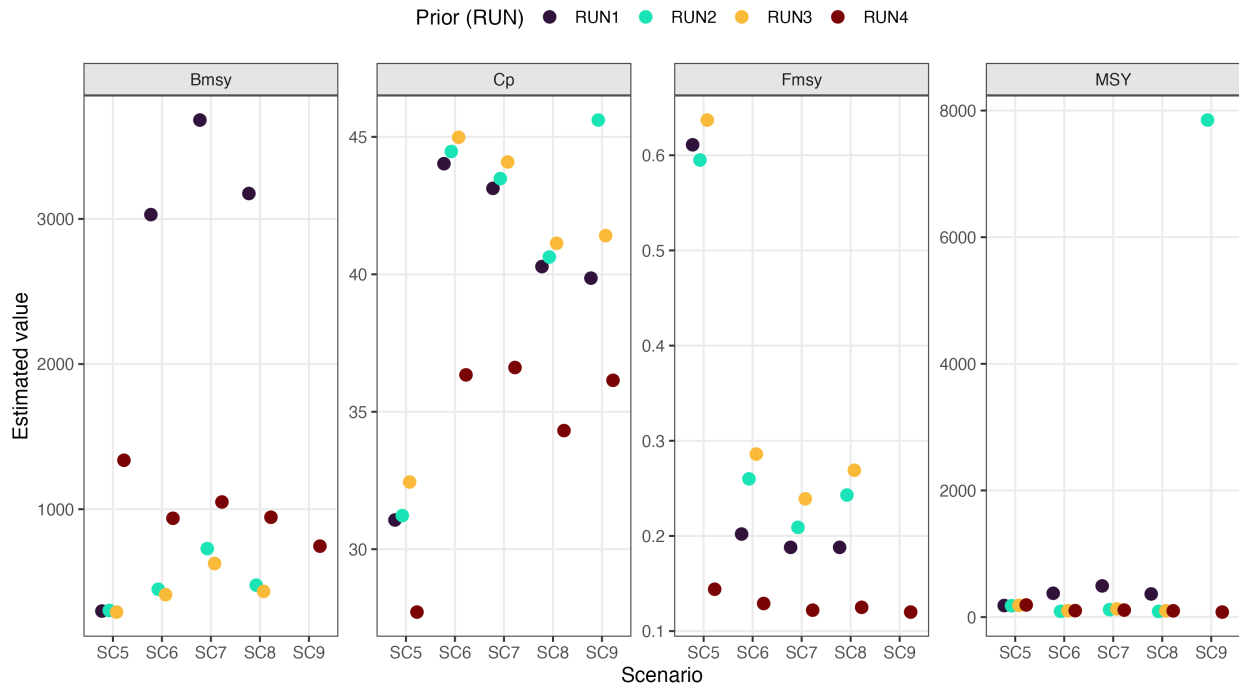


Figure 6: Comparison of stochastic biological reference points (Bmsy, Fmsy, MSY) and current catch (Cp) across SPiCT Scenarios SC5–SC9 and prior configurations (RUN1–RUN4) for Nephrops FU30. Points represent scenario–prior combinations, illustrating the sensitivity of stock productivity and exploitation reference points to alternative data configurations and prior assumptions.

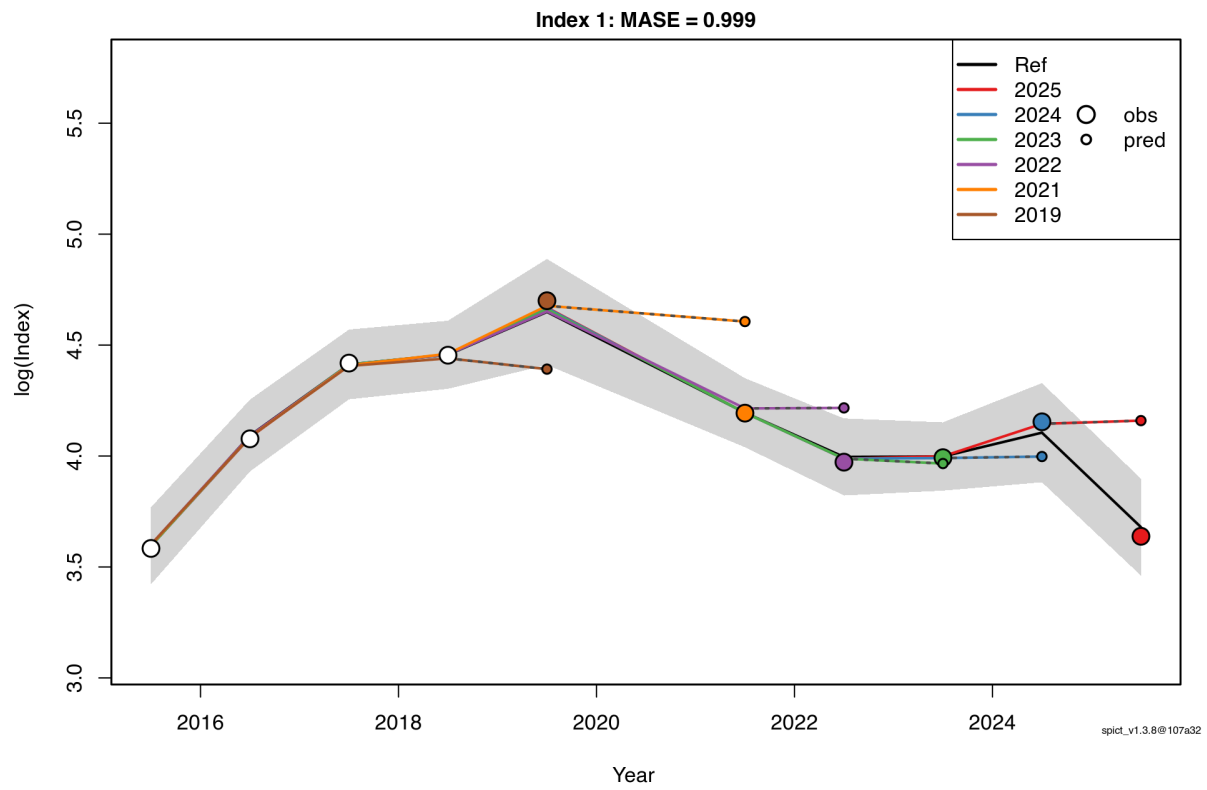


Figure 7: Hindcast diagnostics for Scenario SC0 under prior configuration RUN4.

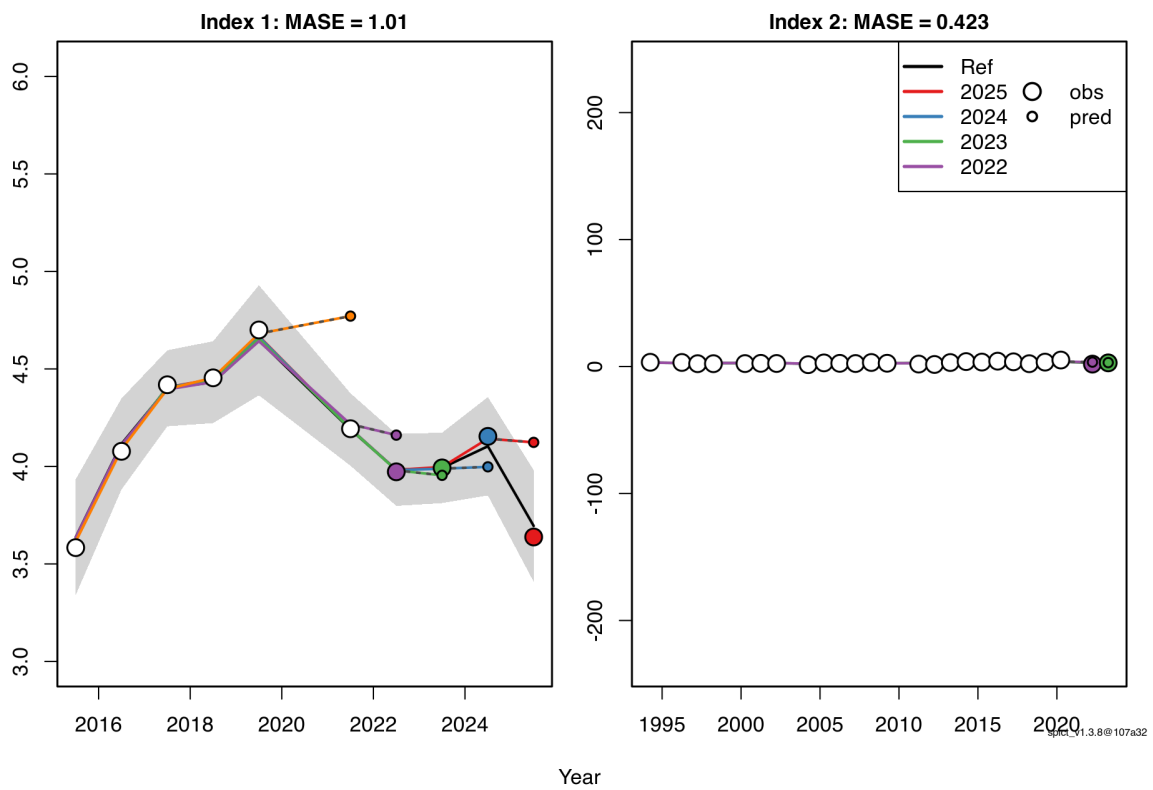


Figure 8: Hindcast diagnostics for Scenario SC1 under prior configuration RUN1.

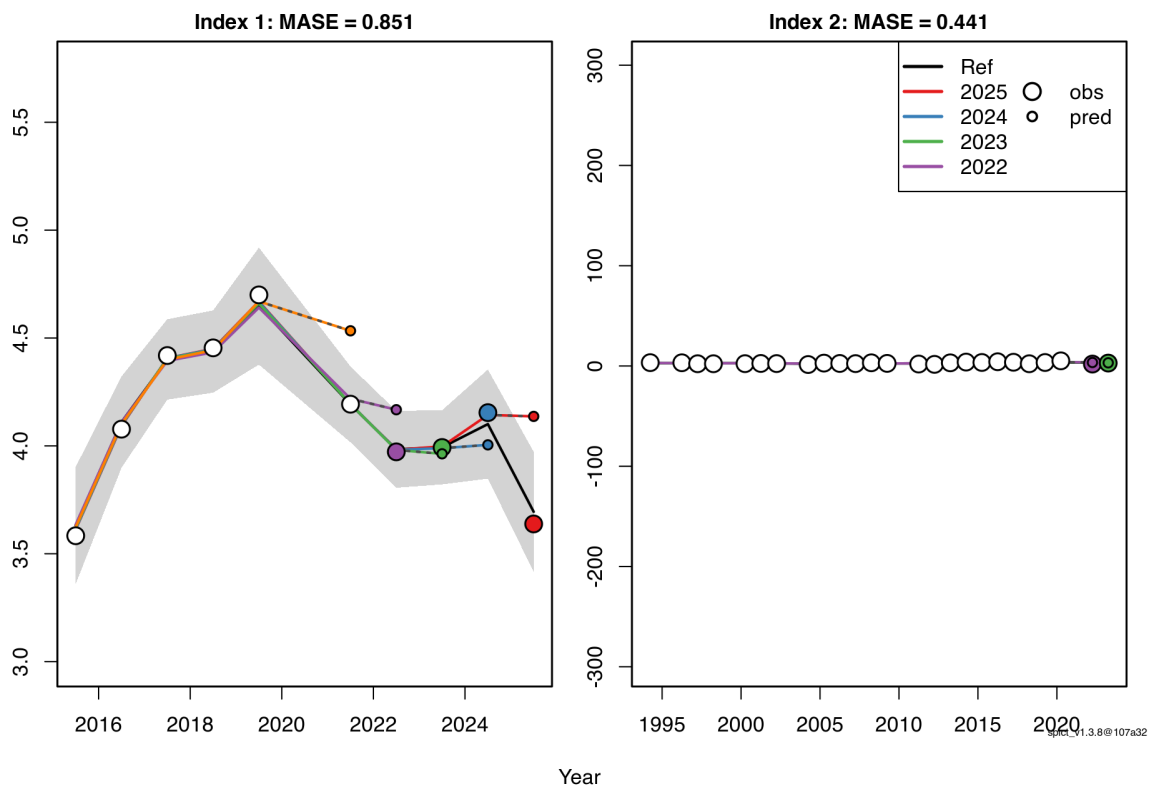


Figure 9: Hindcast diagnostics for Scenario SC1 under prior configuration RUN2.

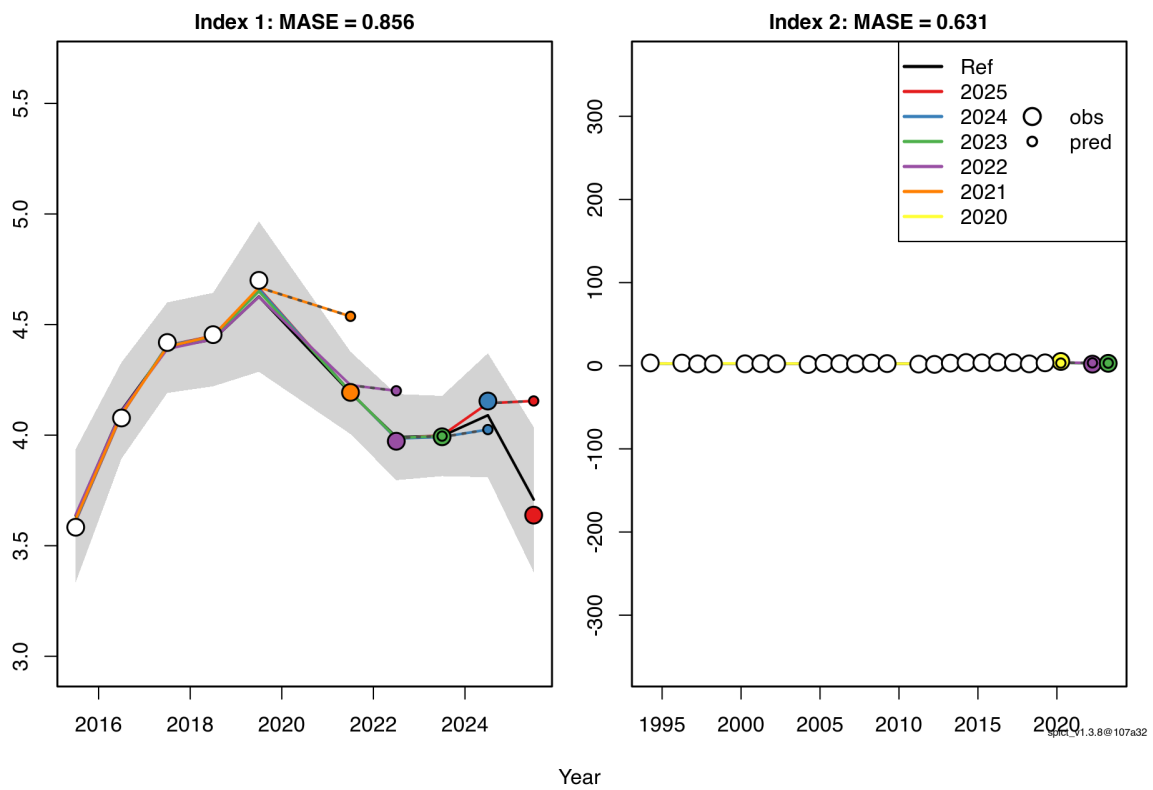


Figure 10: Hindcast diagnostics for Scenario SC1 under prior configuration RUN3.

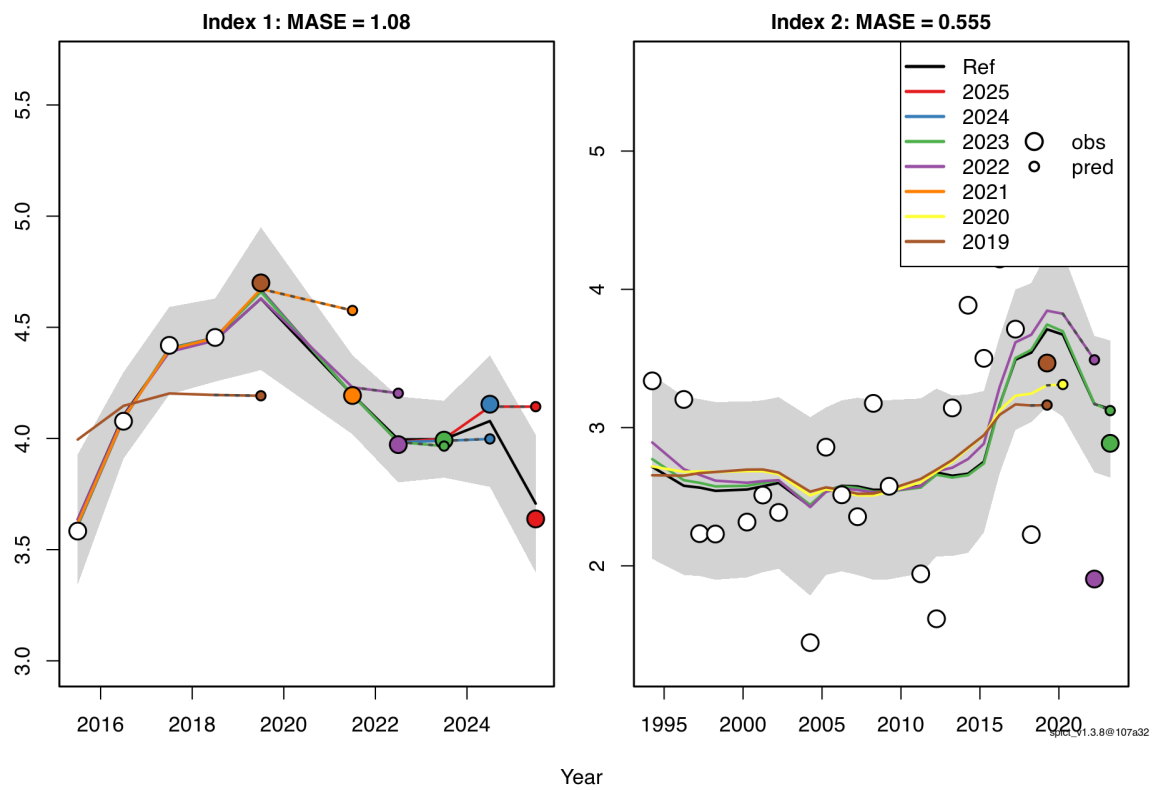


Figure 11: Hindcast diagnostics for Scenario SC1 under prior configuration RUN4.

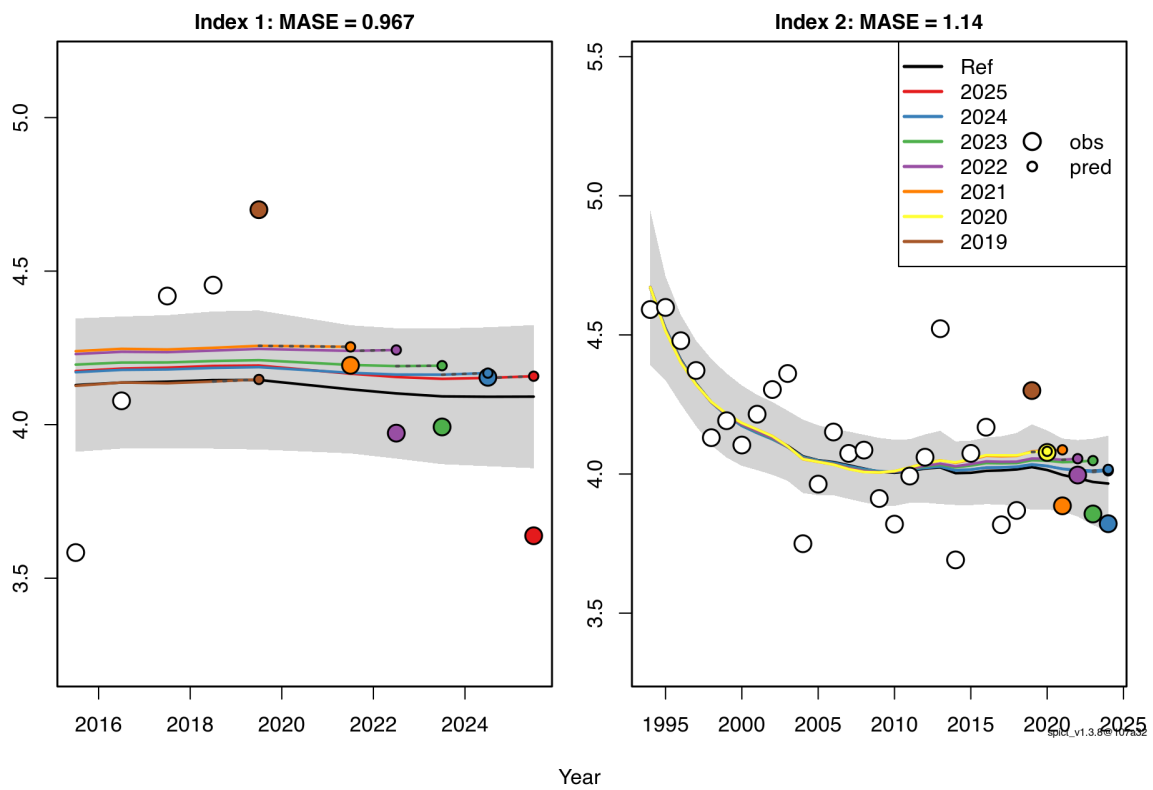


Figure 12: Hindcast diagnostics for Scenario SC2 under prior configuration RUN1.

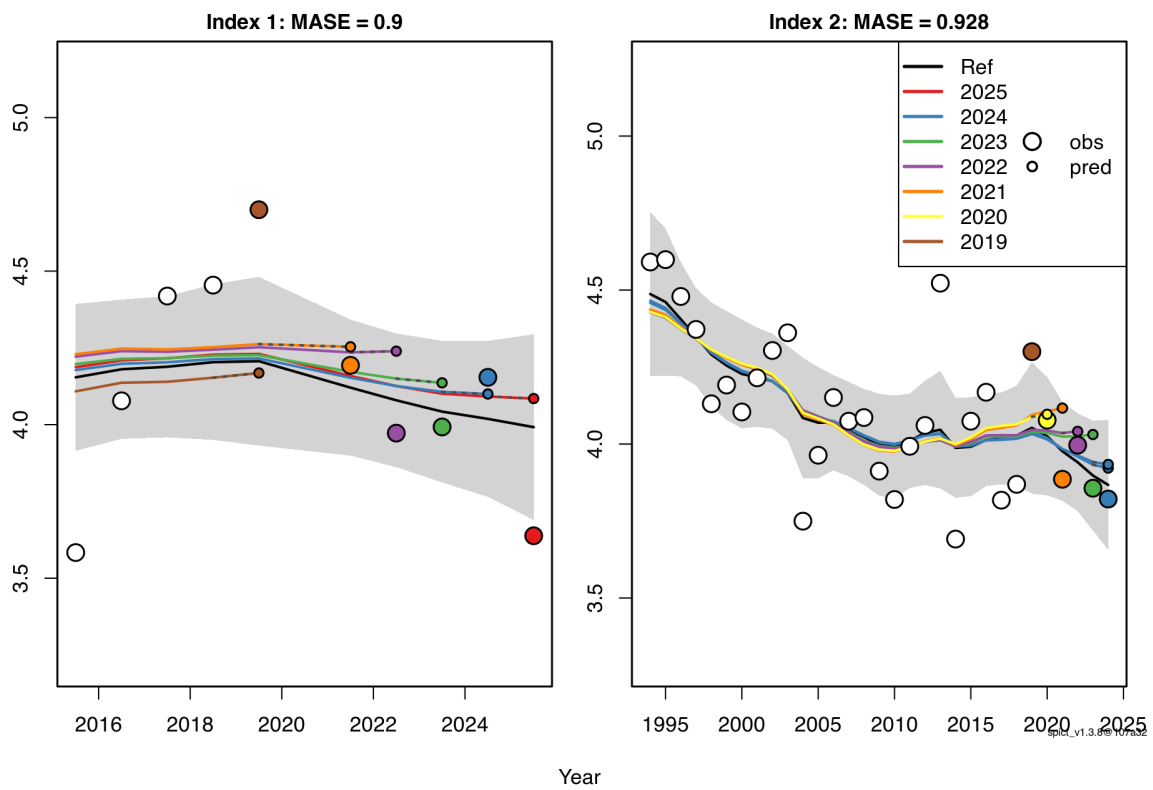


Figure 13: Hindcast diagnostics for Scenario SC2 under prior configuration RUN2.

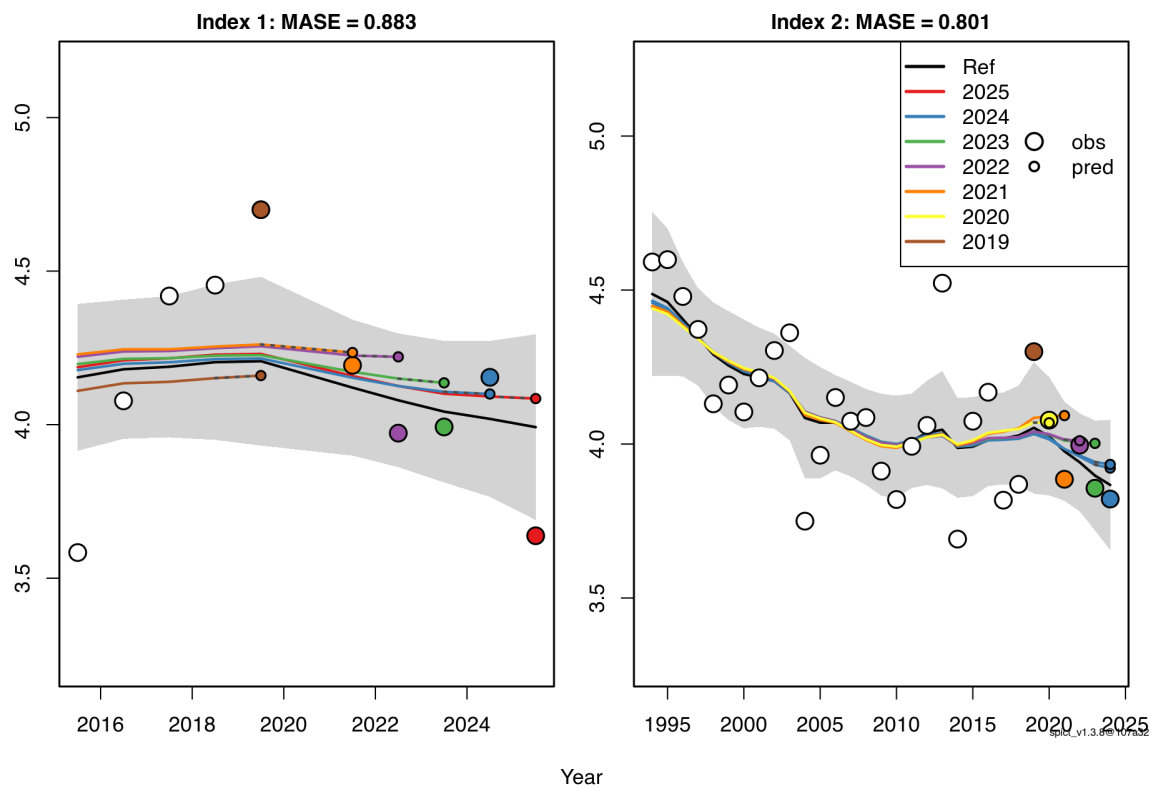


Figure 14: Hindcast diagnostics for Scenario SC2 under prior configuration RUN3.

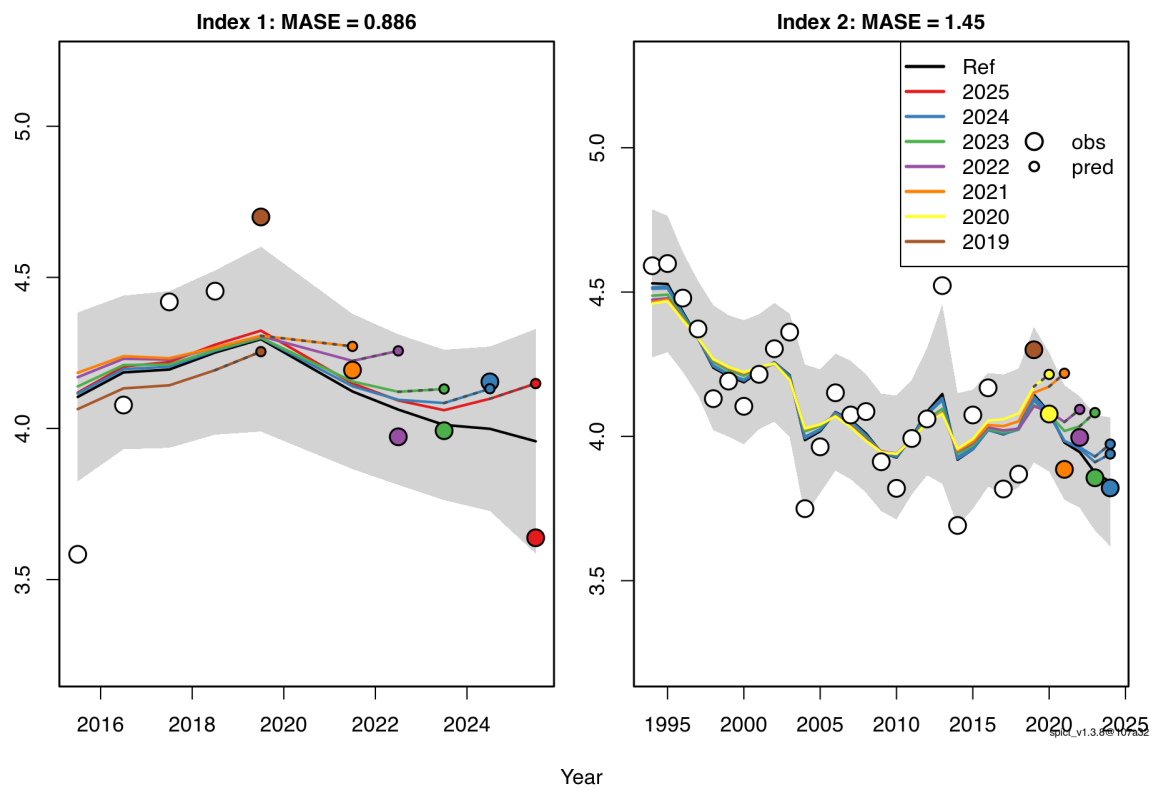
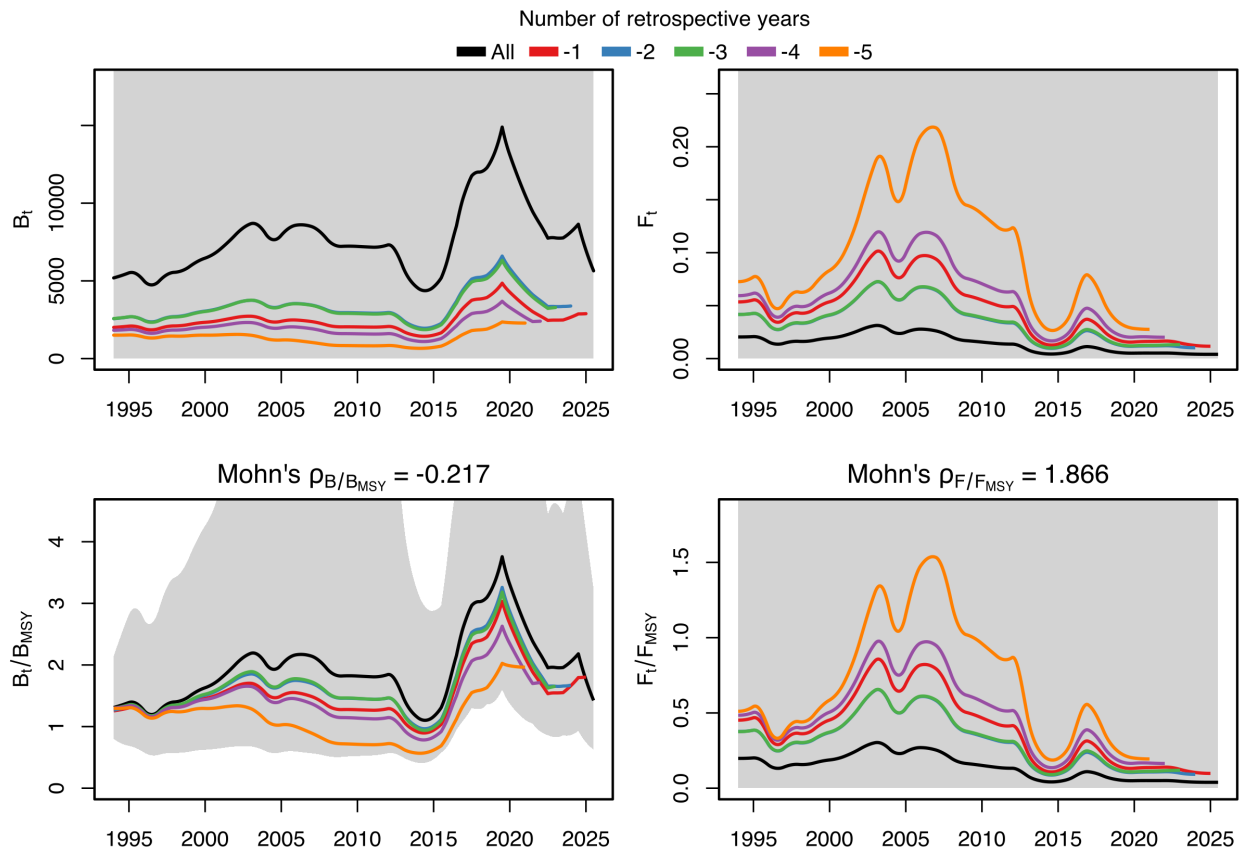
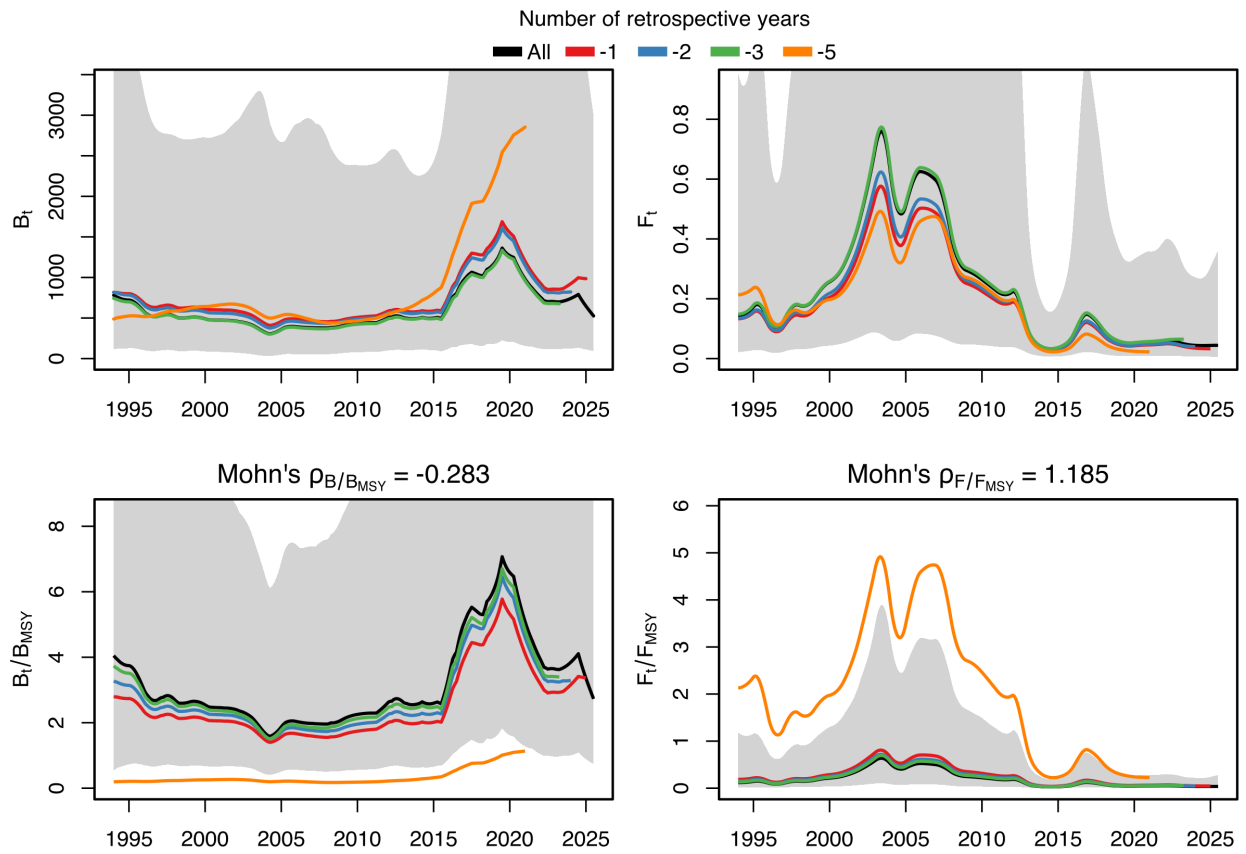


Figure 15: Hindcast diagnostics for Scenario SC2 under prior configuration RUN4.



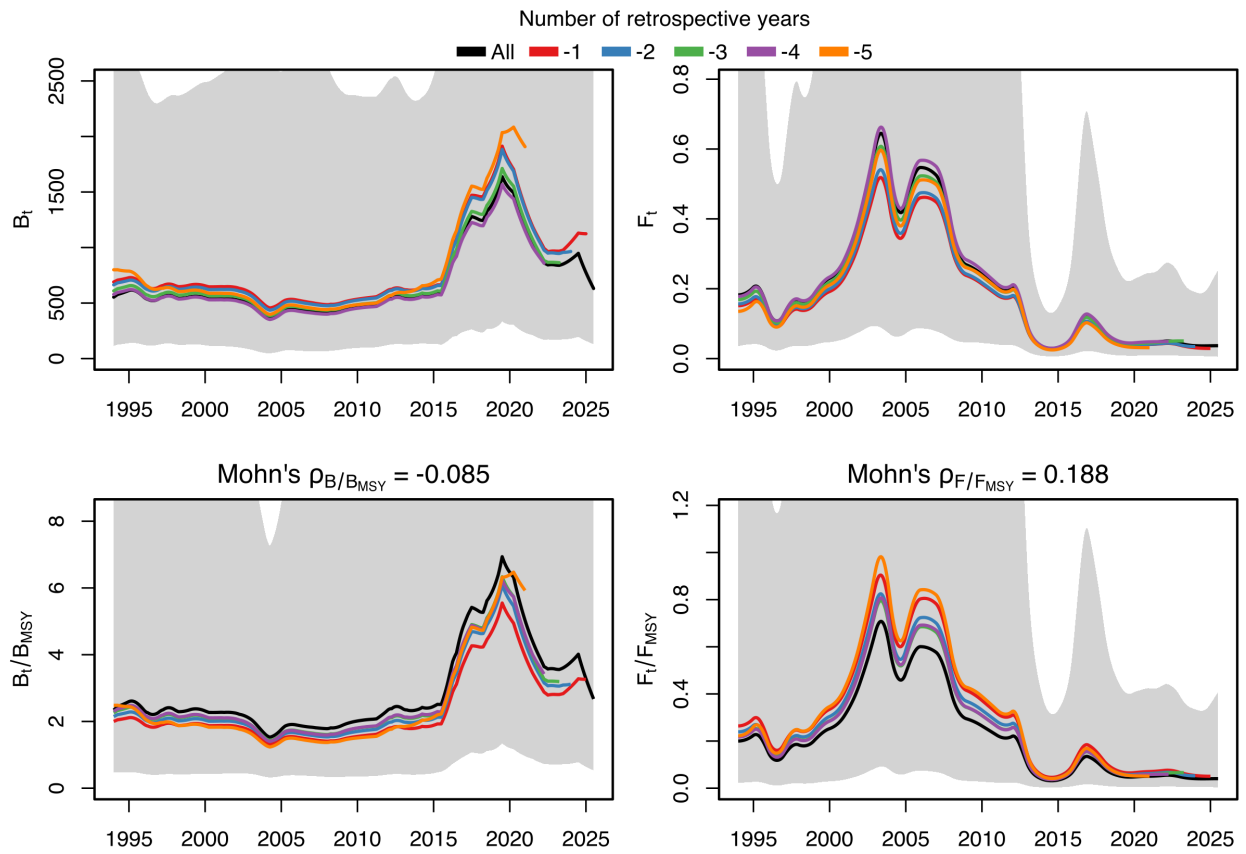
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Figure 16: Retrospective analysis diagnostics for Scenario SC0 under prior configuration RUN4.



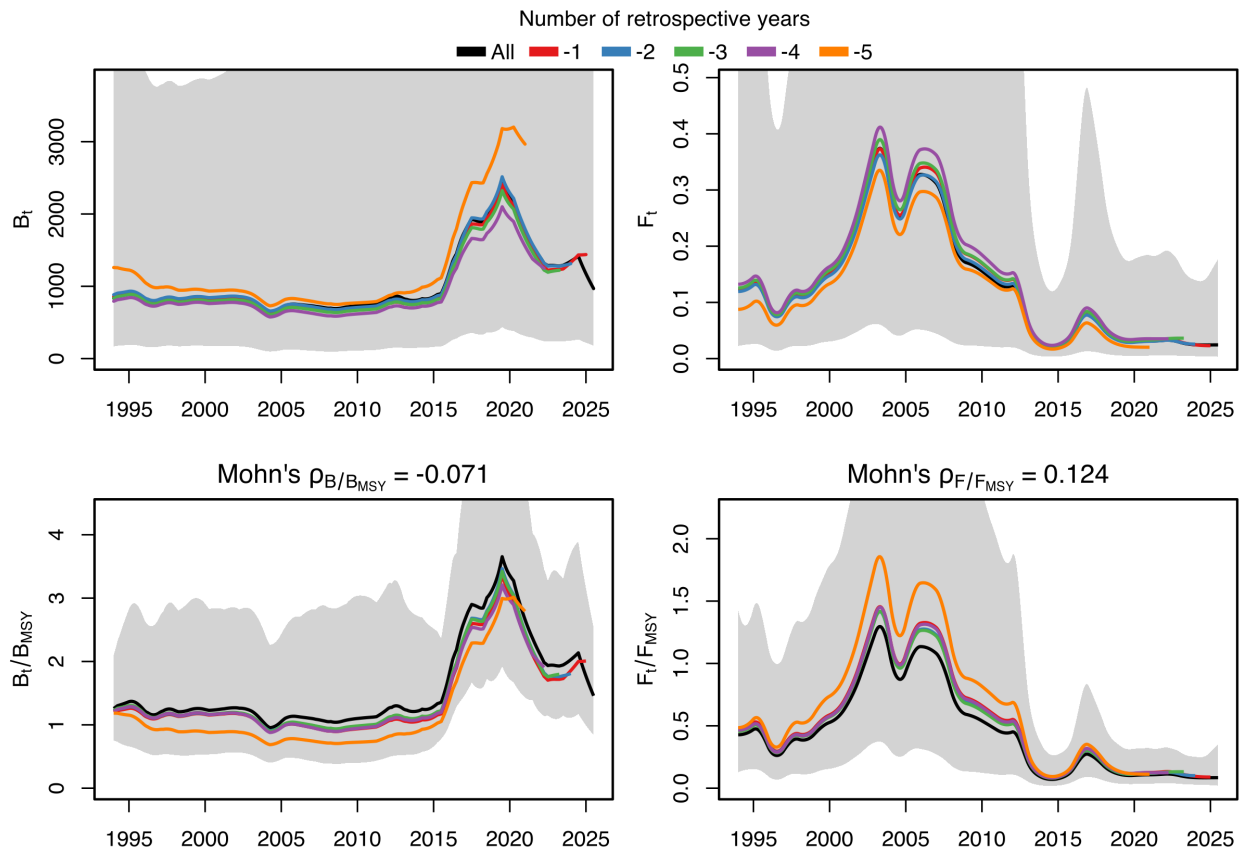
spict_v1.3.8@107a32

Figure 17: Retrospective analysis diagnostics for Scenario SC1 under prior configuration RUN1.



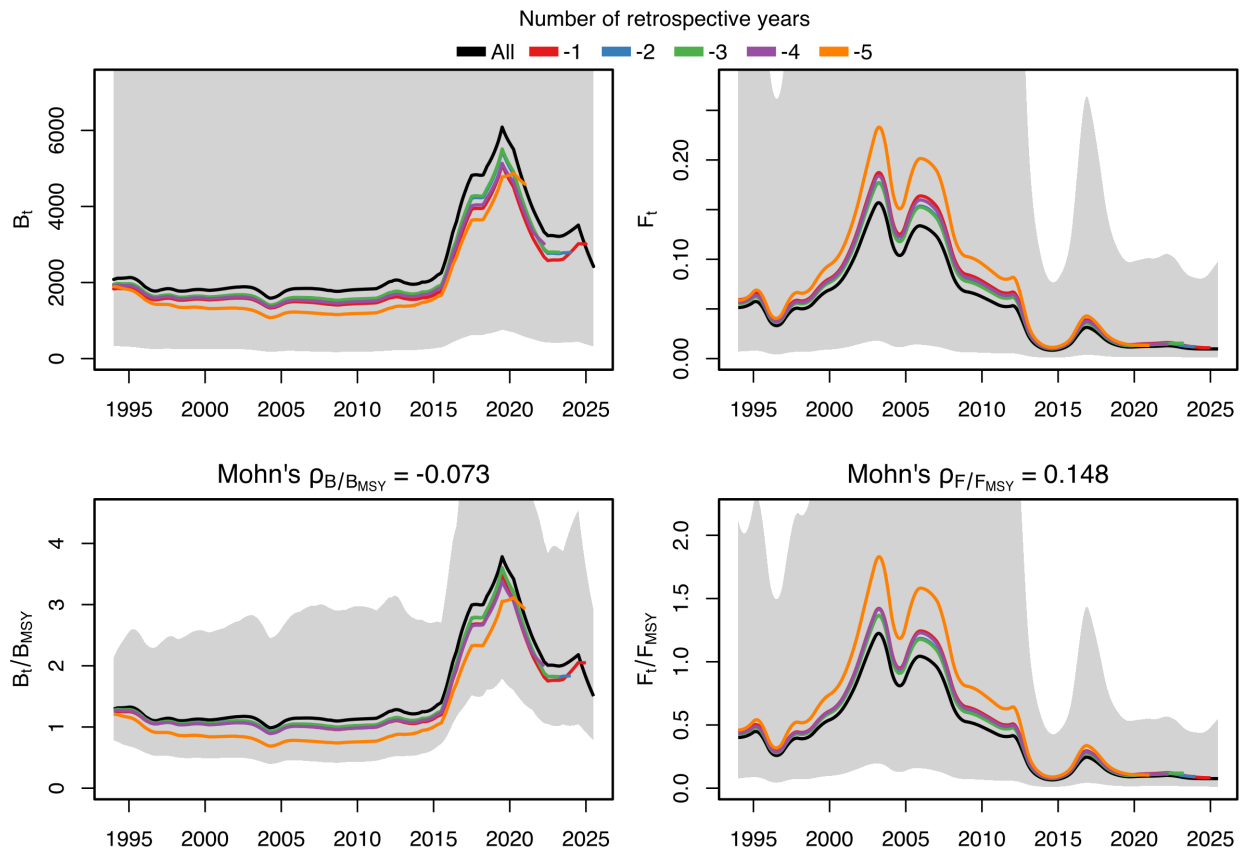
spict_v1.3.8@107a32

Figure 18: Retrospective analysis diagnostics for Scenario SC1 under prior configuration RUN2.



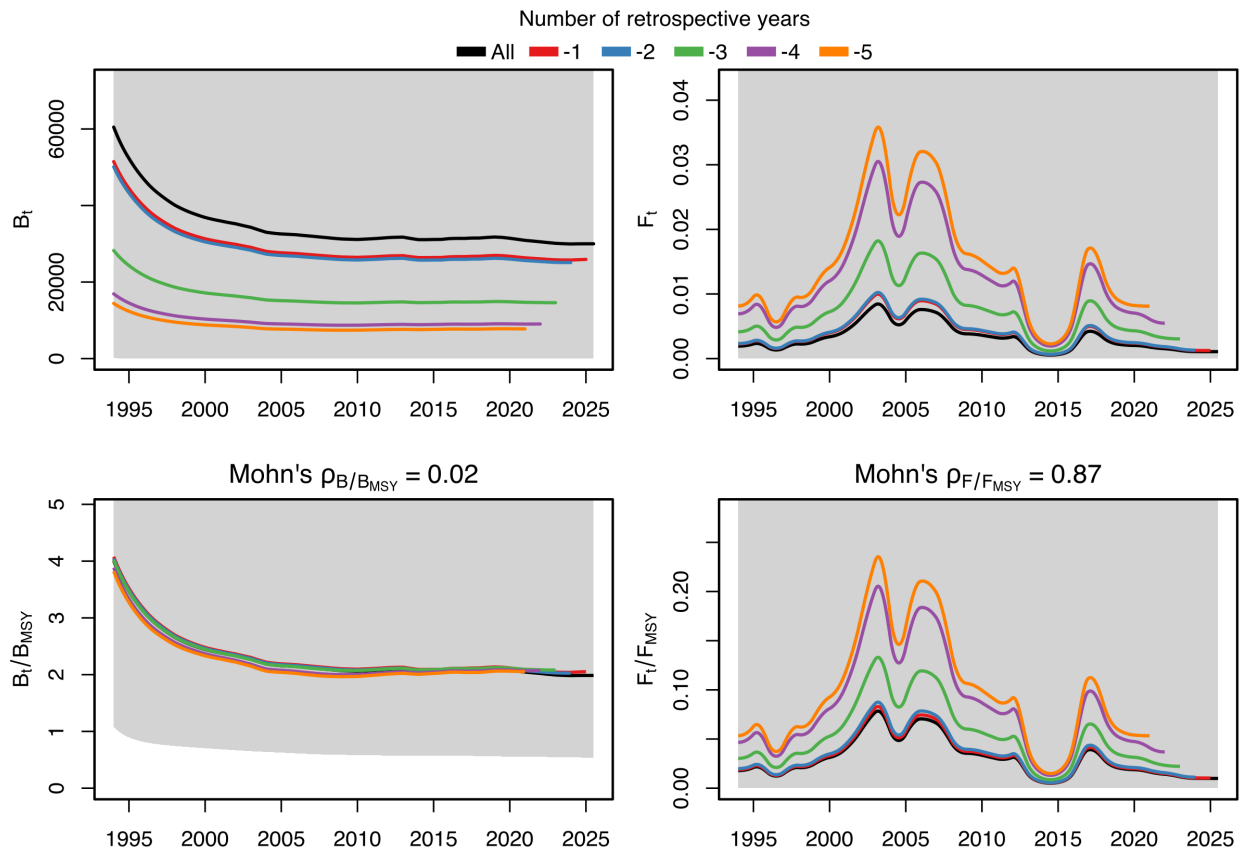
spict_v1.3.8@107a32

Figure 19: Retrospective analysis diagnostics for Scenario SC1 under prior configuration RUN3.



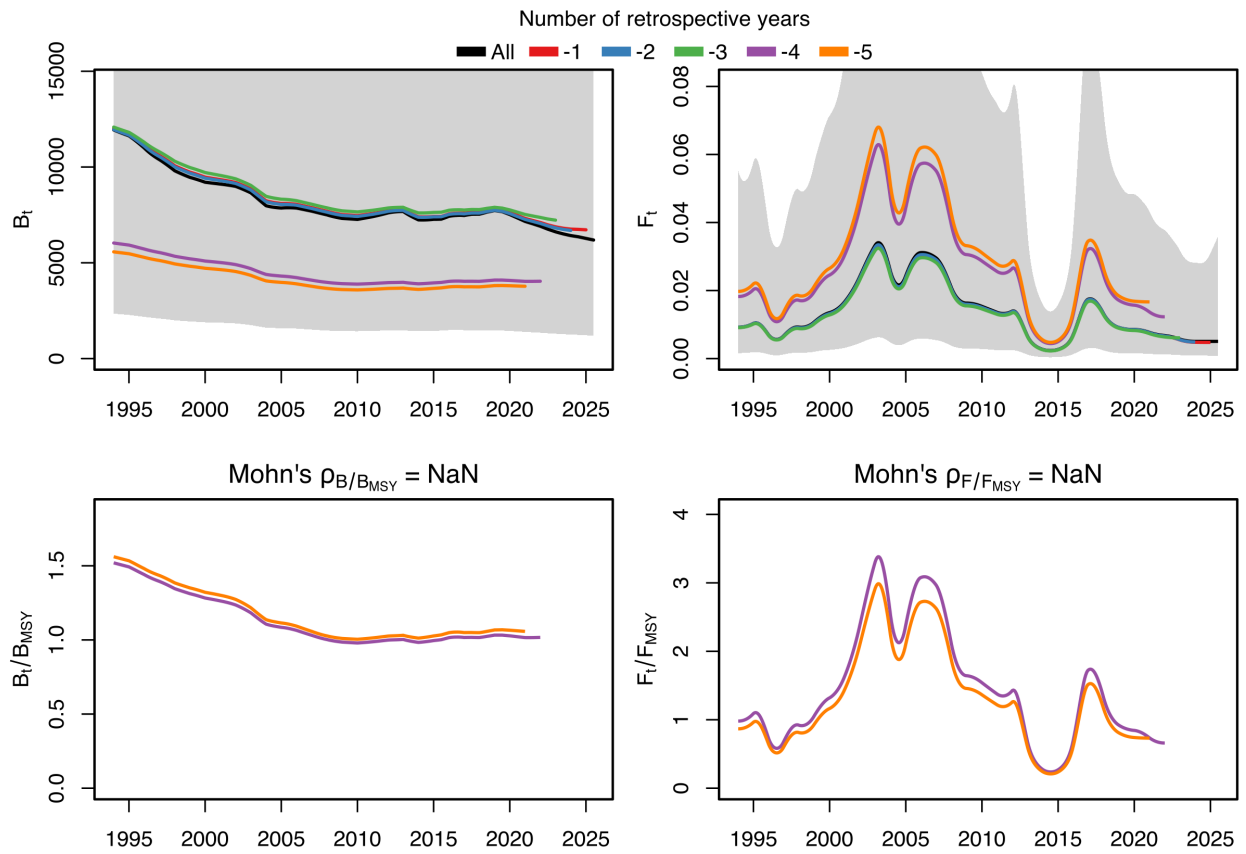
spict_v1.3.8@107a32

Figure 20: Retrospective analysis diagnostics for Scenario SC1 under prior configuration RUN4.



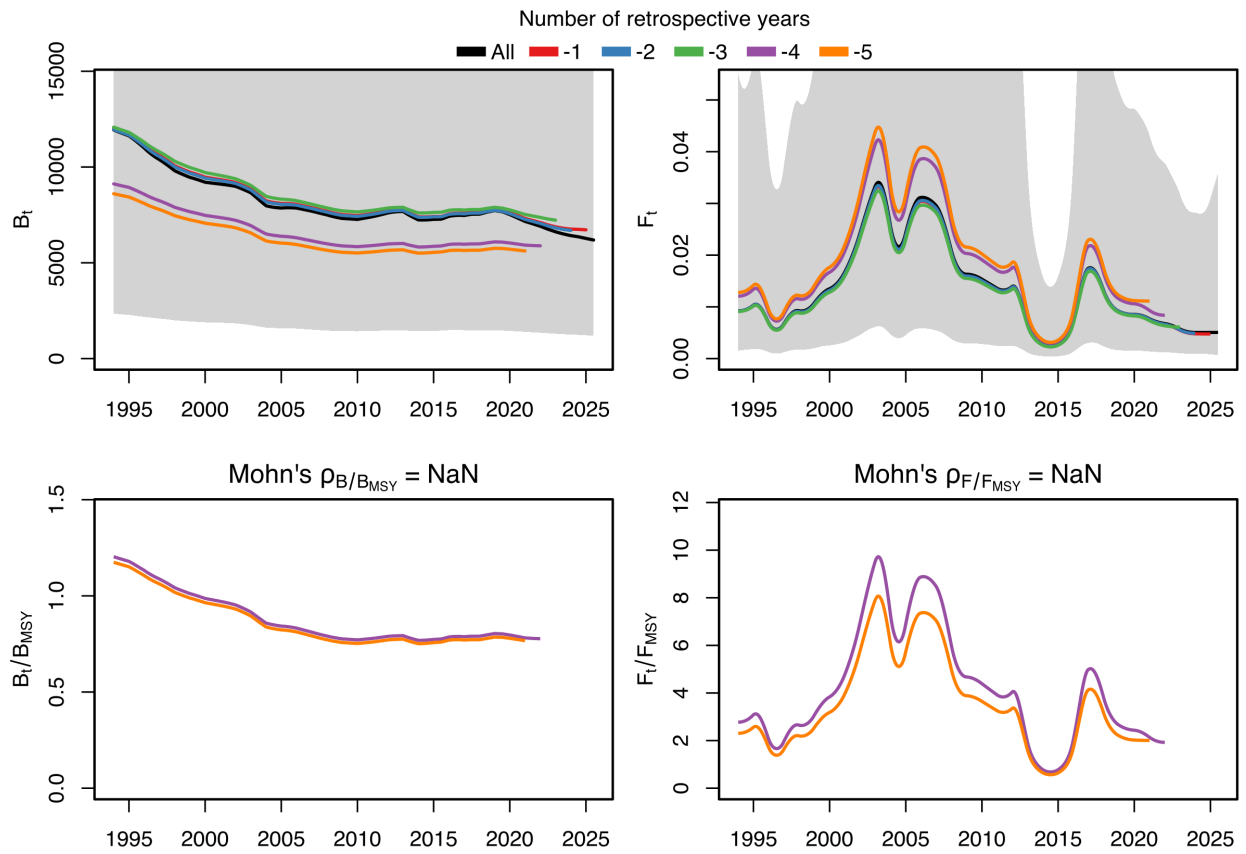
spict_v1.3.8@107a32

Figure 21: Retrospective analysis diagnostics for Scenario SC2 under prior configuration RUN1.



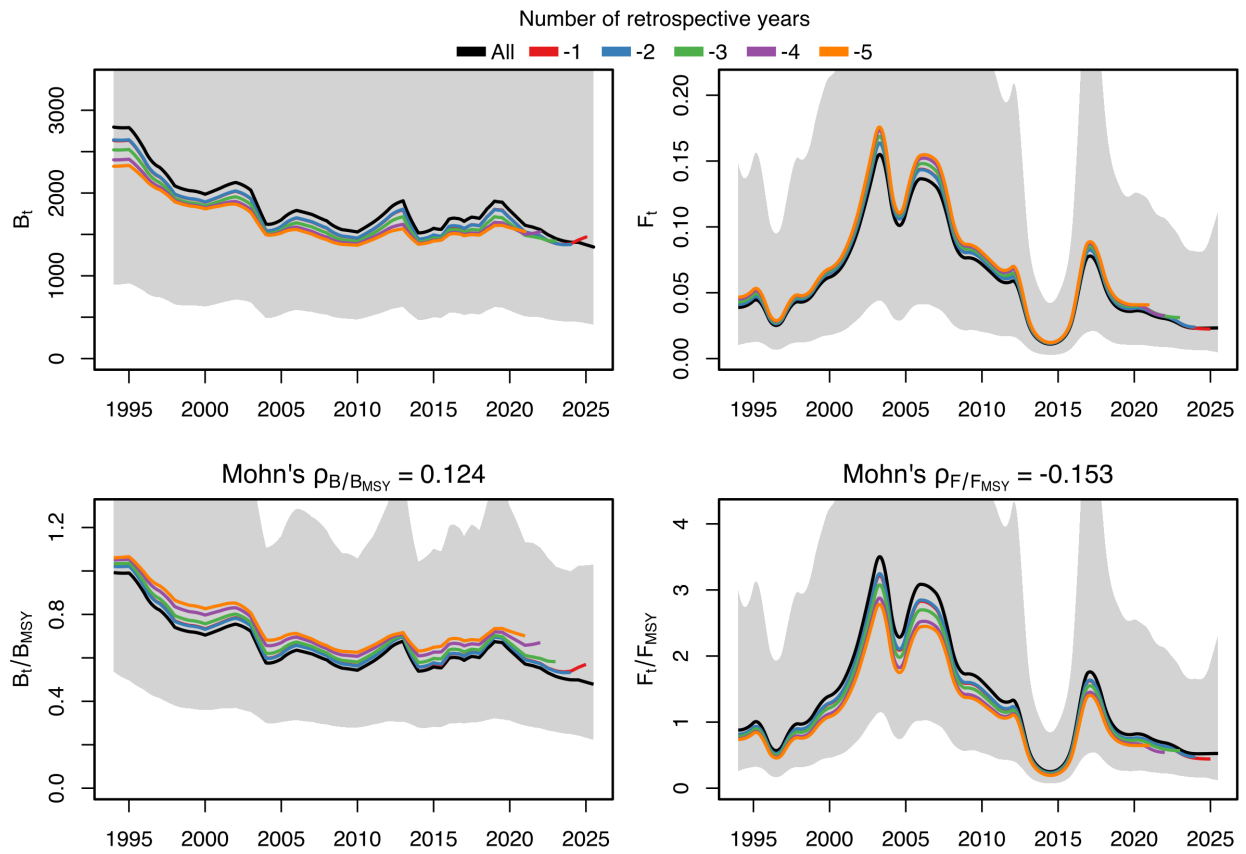
spict_v1.3.8@107a32

Figure 22: Retrospective analysis diagnostics for Scenario SC2 under prior configuration RUN2.



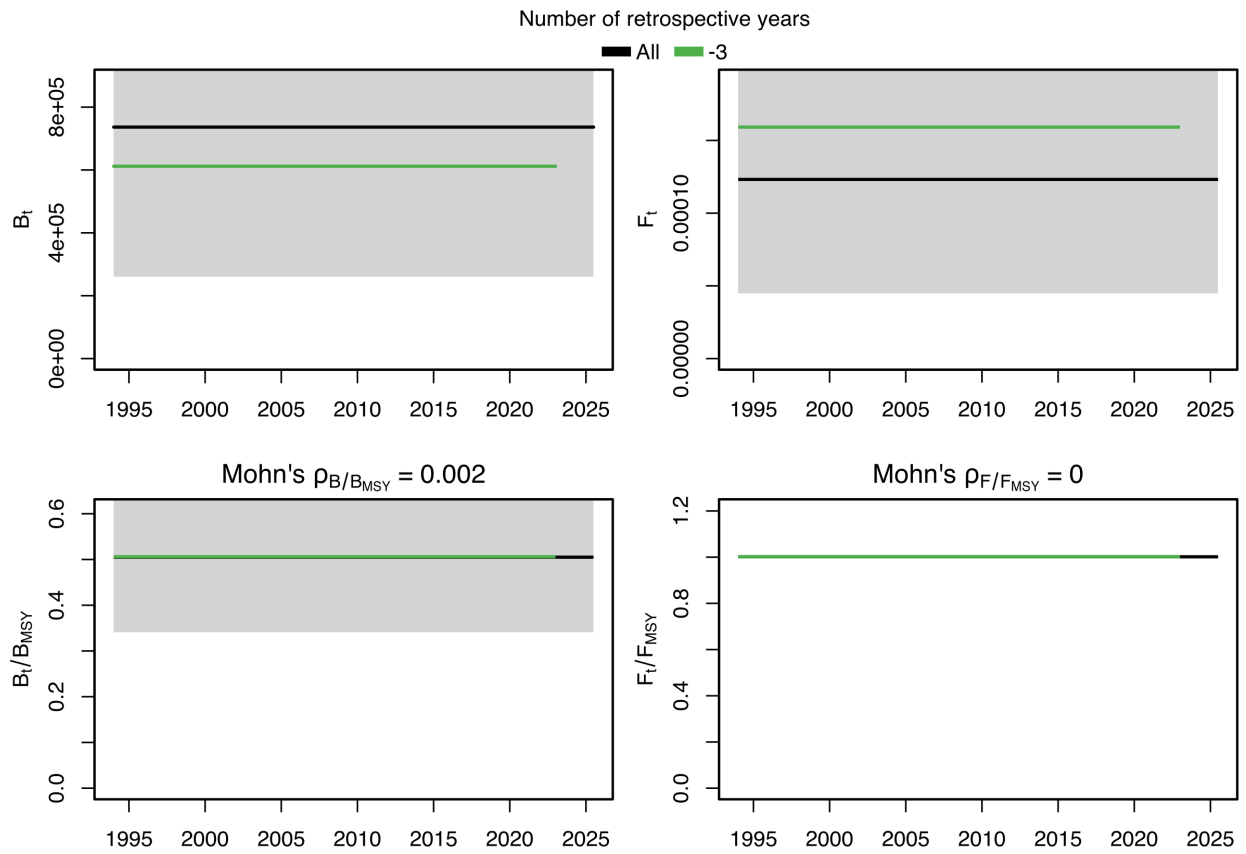
spict_v1.3.8@107a32

Figure 23: Retrospective analysis diagnostics for Scenario SC2 under prior configuration RUN3.



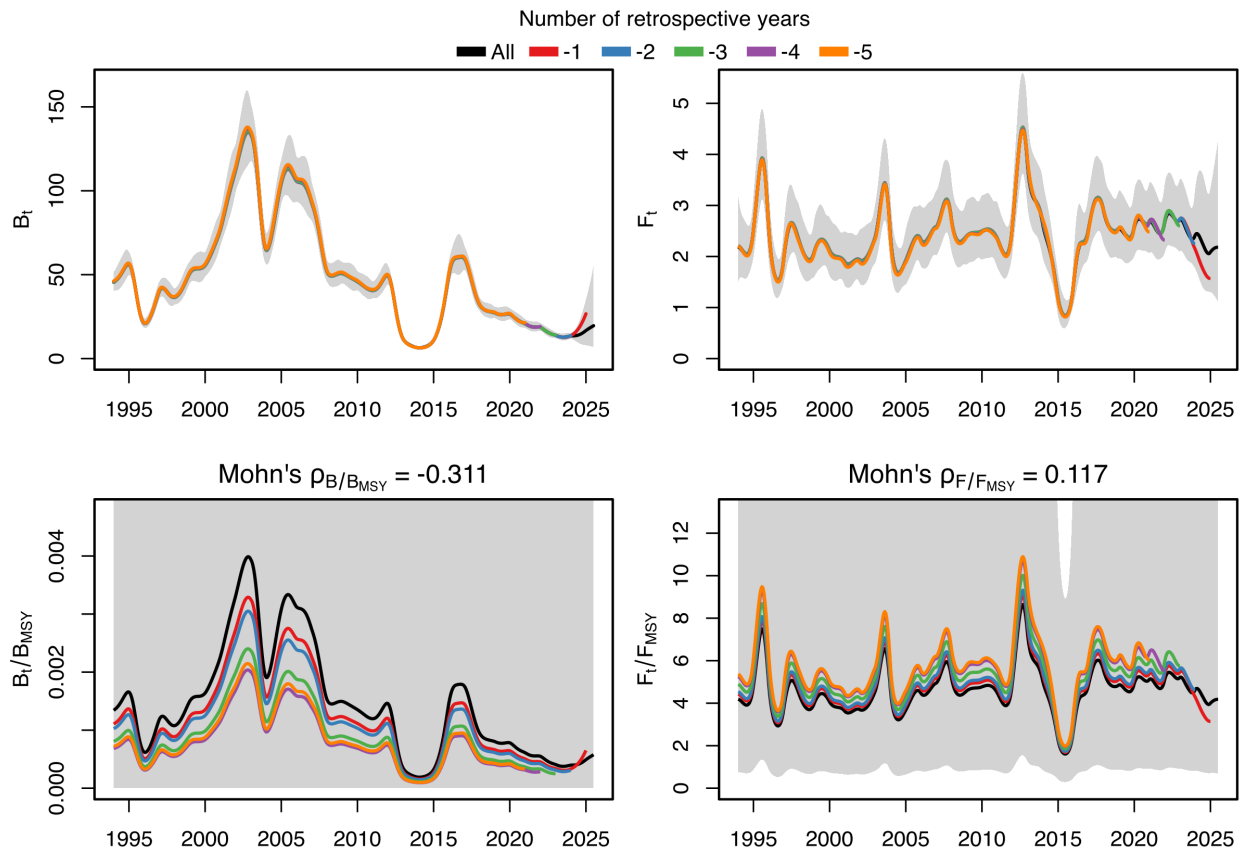
spict_v1.3.8@107a32

Figure 24: Retrospective analysis diagnostics for Scenario SC2 under prior configuration RUN4.



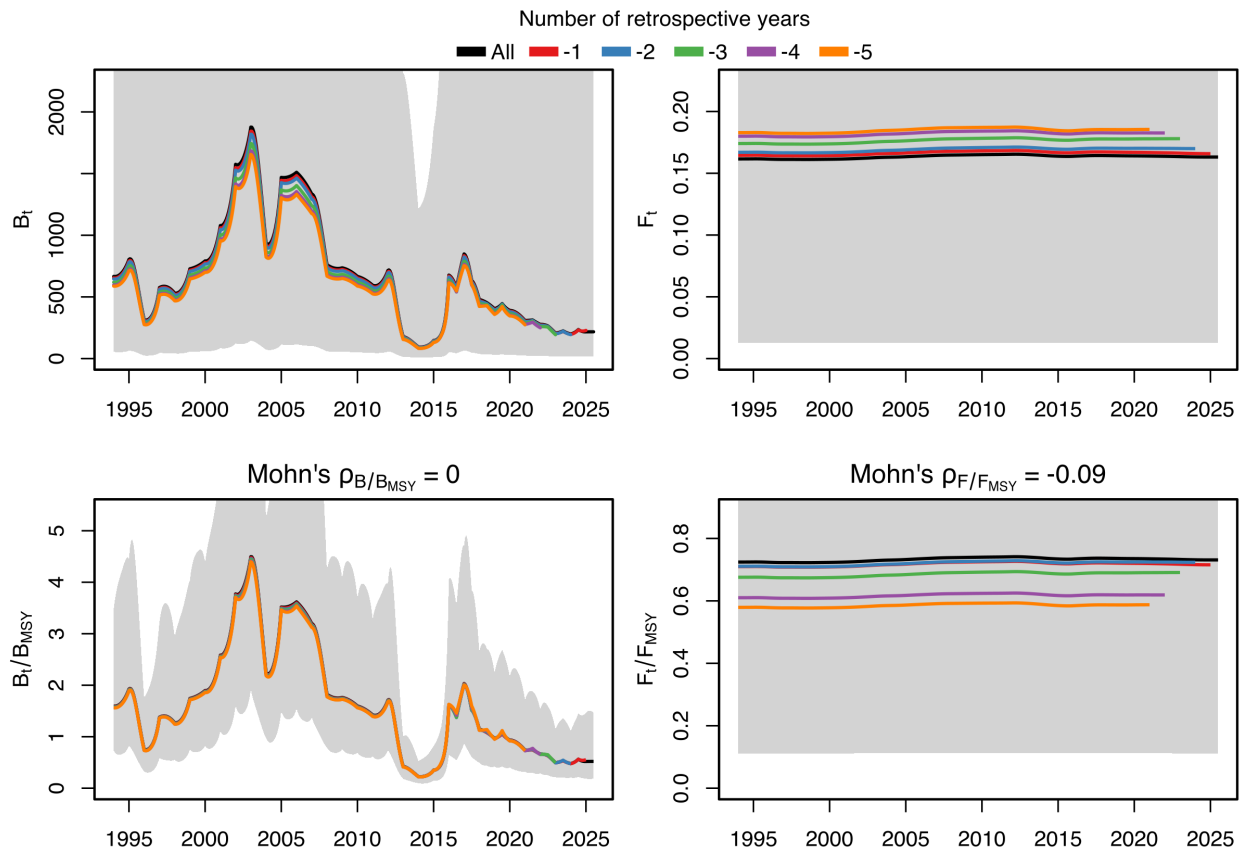
spict_v1.3.8@107a32

Figure 25: Retrospective analysis diagnostics for Scenario SC3 under prior configuration RUN4.



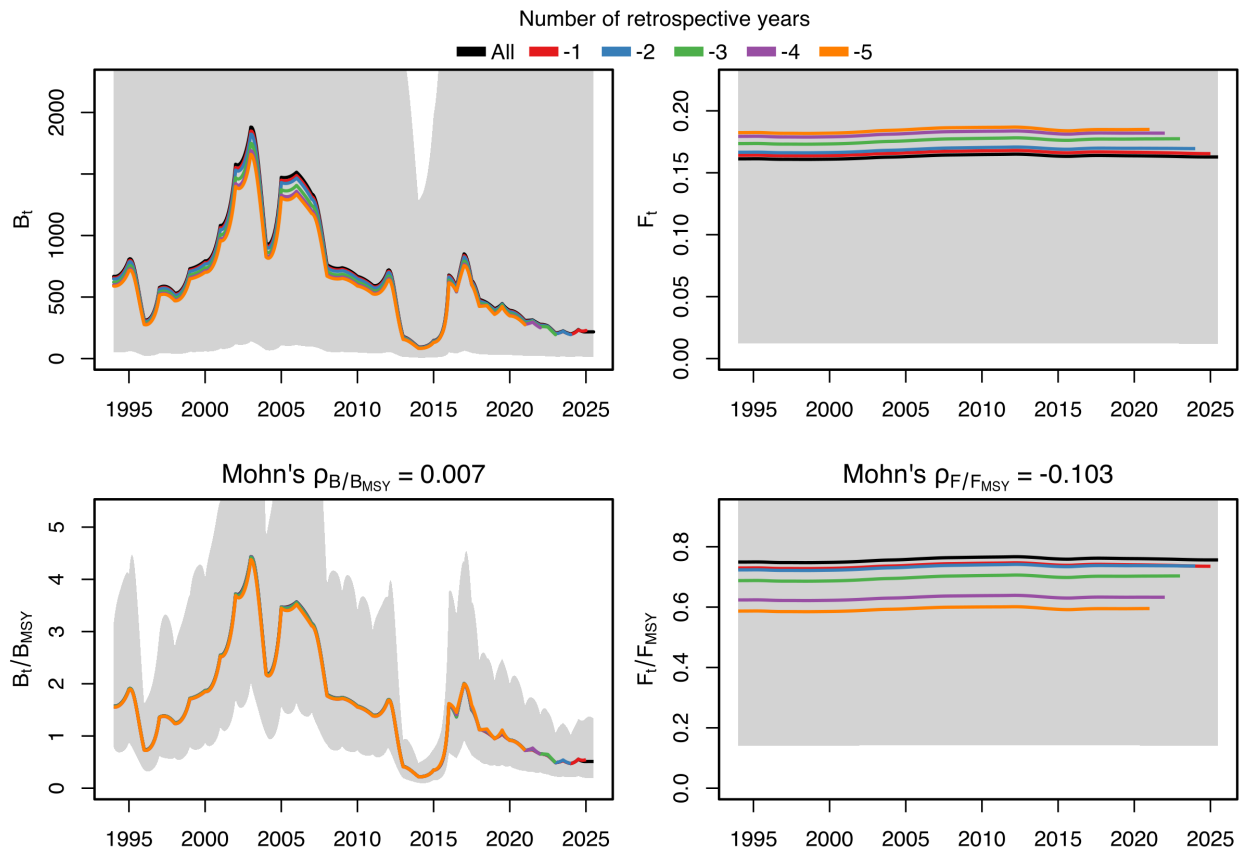
spict_v1.3.8@107a32

Figure 26: Retrospective analysis diagnostics for Scenario SC4 under prior configuration RUN1.



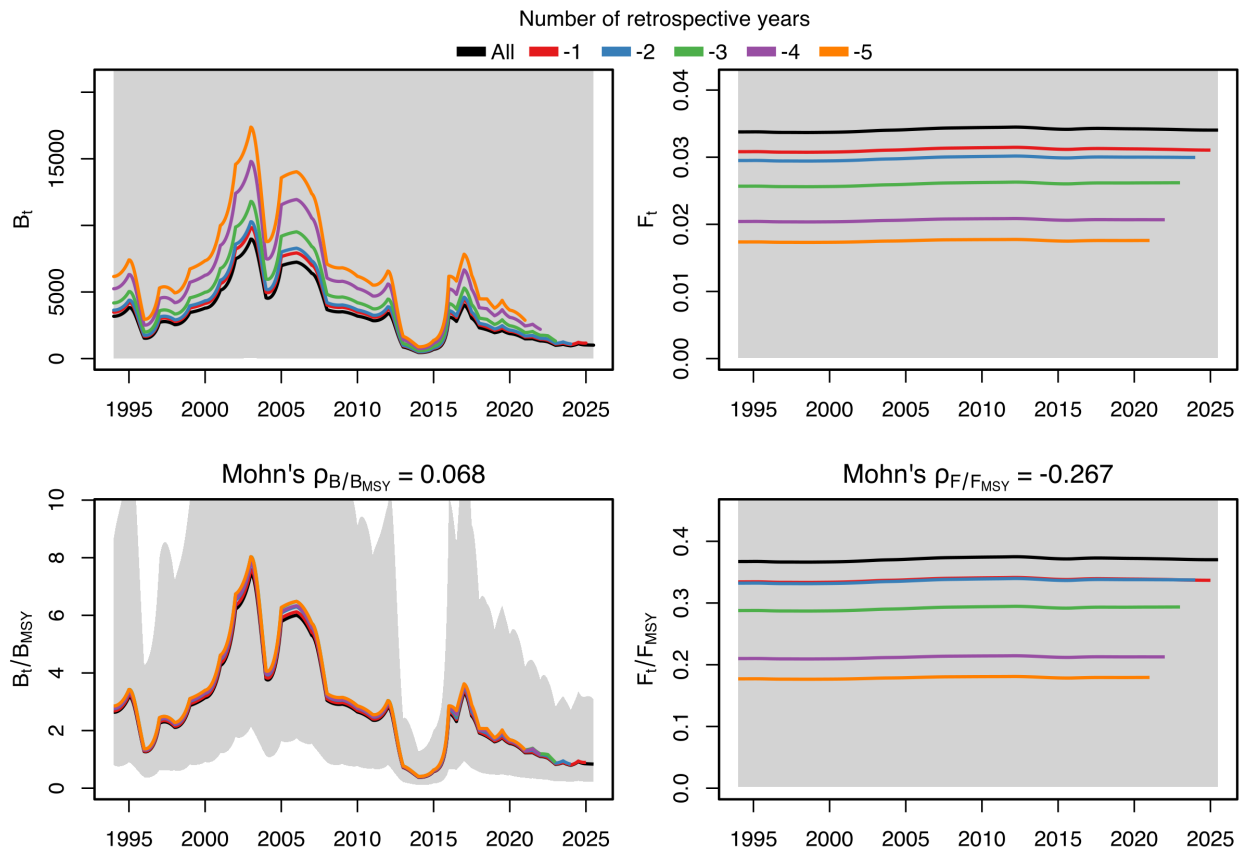
spict_v1.3.8@107a32

Figure 27: Retrospective analysis diagnostics for Scenario SC4 under prior configuration RUN2.



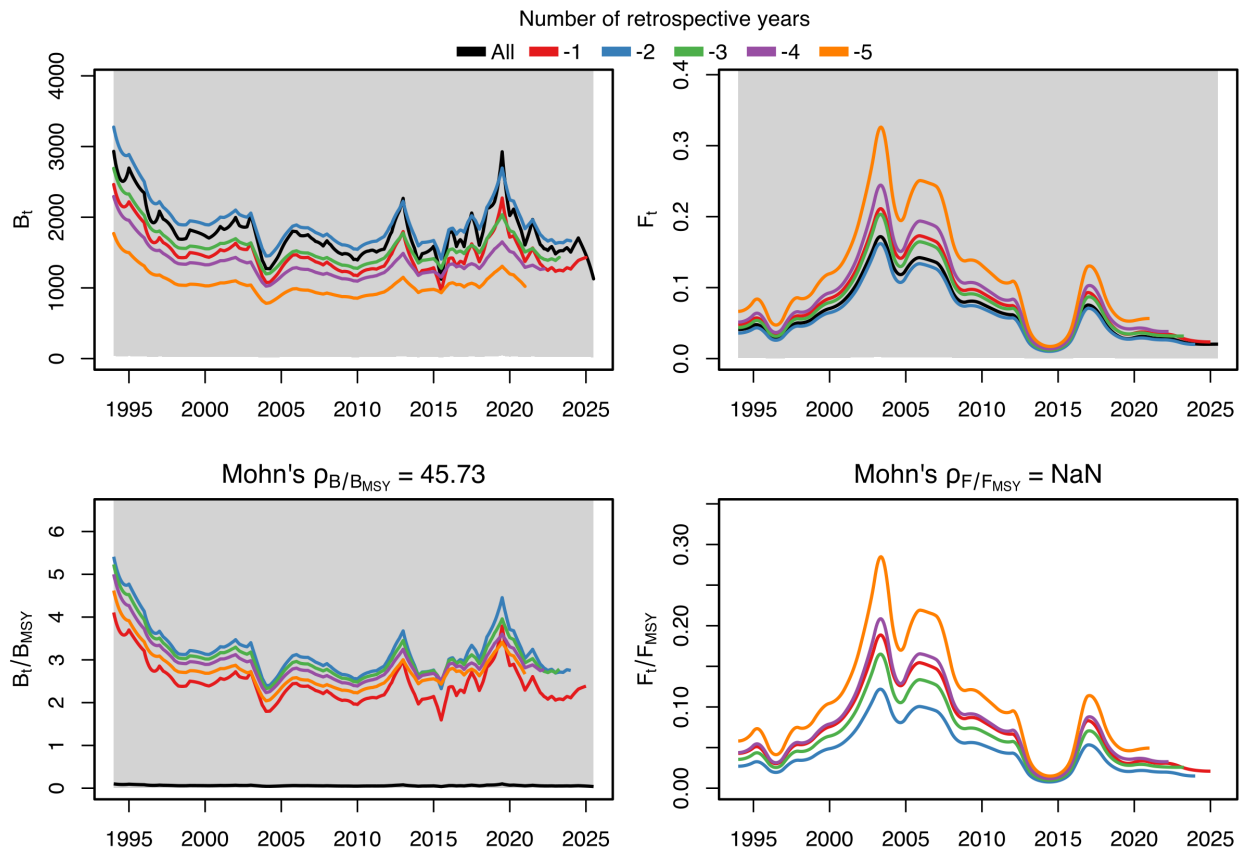
spict_v1.3.8@107a32

Figure 28: Retrospective analysis diagnostics for Scenario SC4 under prior configuration RUN3.



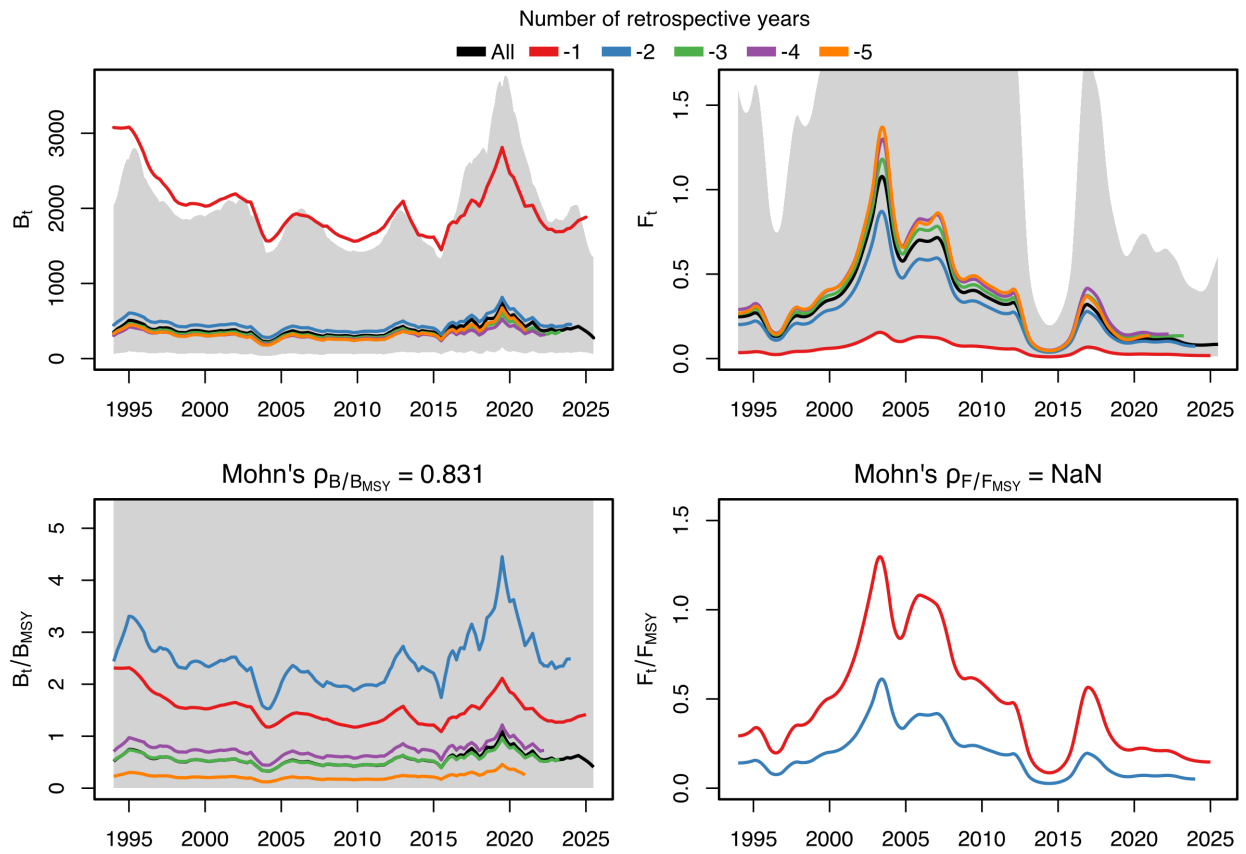
spict_v1.3.8@107a32

Figure 29: Retrospective analysis diagnostics for Scenario SC4 under prior configuration RUN4.



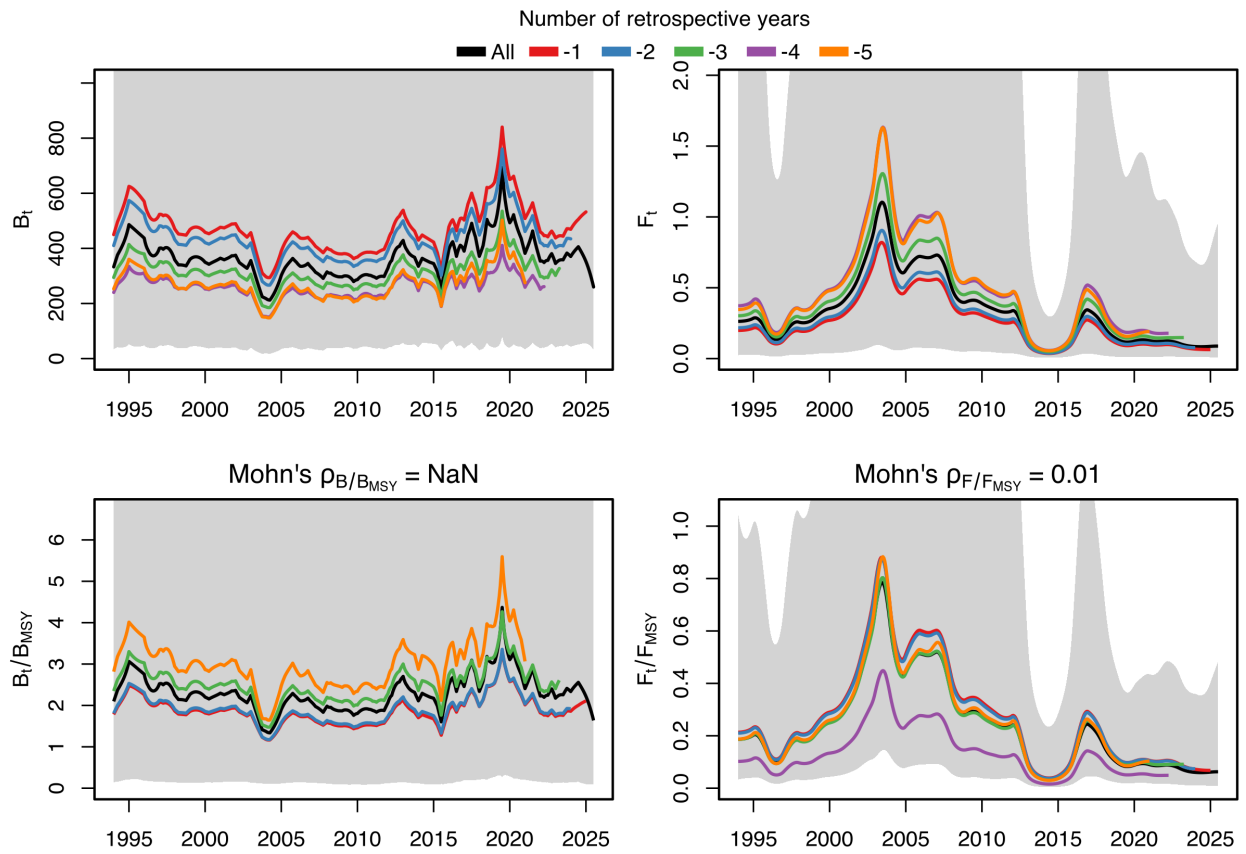
spict_v1.3.8@107a32

Figure 30: Retrospective analysis diagnostics for Scenario SC5 under prior configuration RUN1.



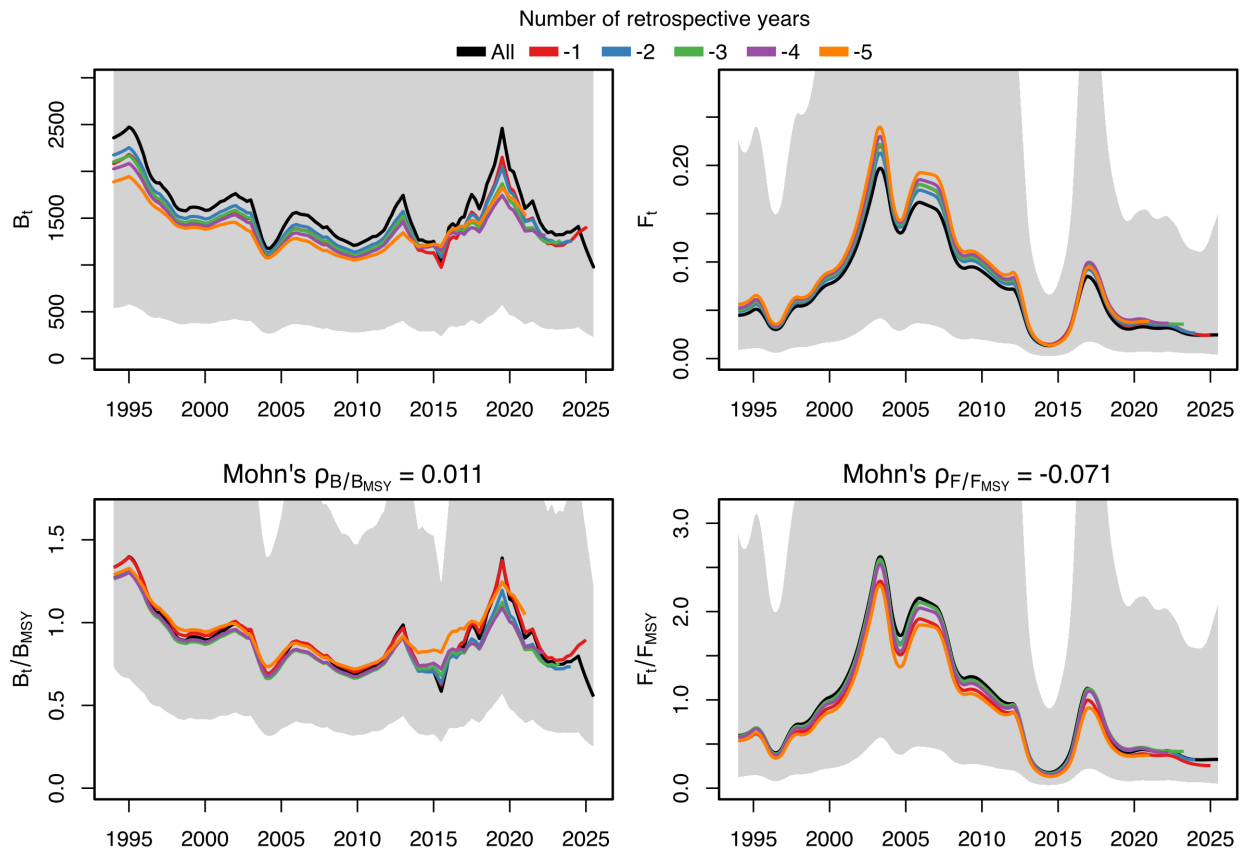
spict_v1.3.8@107a32

Figure 31: Retrospective analysis diagnostics for Scenario SC5 under prior configuration RUN2.



spict_v1.3.8@107a32

Figure 32: Retrospective analysis diagnostics for Scenario SC5 under prior configuration RUN3.



spict_v1.3.8@107a32

Figure 33: Retrospective analysis diagnostics for Scenario SC5 under prior configuration RUN4.

Table 6: Stochastic biological reference points and estimated current catch (Cp) by SPiCT scenario and prior configuration (RUN) for Nephrops FU30.

Scenario	Run	$B_{\{MSY\}}$	$F_{\{MSY\}}$	MSY	Cp
SC0	RUN1	551.406	0.276	151.027	28.099
SC0	RUN2	589.548	0.293	171.299	28.898
SC0	RUN3	603.704	0.284	169.102	28.781
SC0	RUN4	3964.471	0.103	393.814	23.973
SC1	RUN1	194.002	1.213	235.274	26.692
SC1	RUN2	242.131	0.907	219.600	26.197
SC1	RUN3	663.834	0.289	190.983	26.511
SC1	RUN4	1609.166	0.128	204.279	24.745
SC2	RUN1	15102.859	0.108	1627.234	32.302
SC2	RUN2	NA	NA	NA	31.078
SC2	RUN3	NA	NA	NA	31.078
SC2	RUN4	2817.064	0.044	120.904	32.674
SC3	RUN1	59940.847	NA	NA	0.893
SC3	RUN2	69.328	6.884	1239.830	33.734
SC3	RUN3	NA	NA	1911.312	32.171
SC3	RUN4	1458135.555	0.000	179.351	90.834
SC4	RUN1	33996.785	0.522	17739.924	55.213
SC4	RUN2	417.520	0.223	78.345	41.586
SC4	RUN3	424.280	0.215	74.045	41.268
SC4	RUN4	1205.914	0.092	46.054	36.394
SC5	RUN1	578.317	1.980	1144.969	32.065
SC5	RUN2	151.642	1.917	290.676	33.190
SC5	RUN3	158.706	1.406	227.578	34.272
SC5	RUN4	1769.054	0.075	126.318	26.033

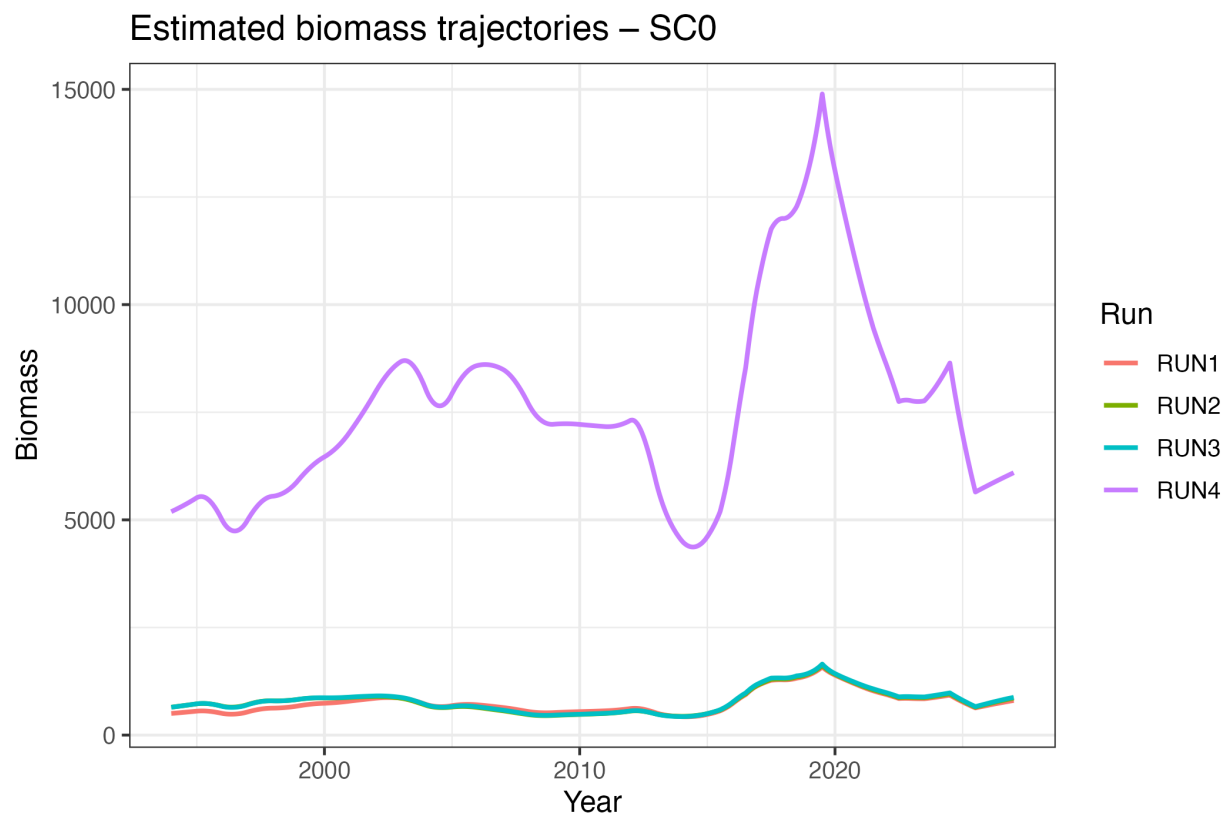


Figure 34: Estimated relative biomass trajectory for Scenario SC0 derived from the SPiCT model.

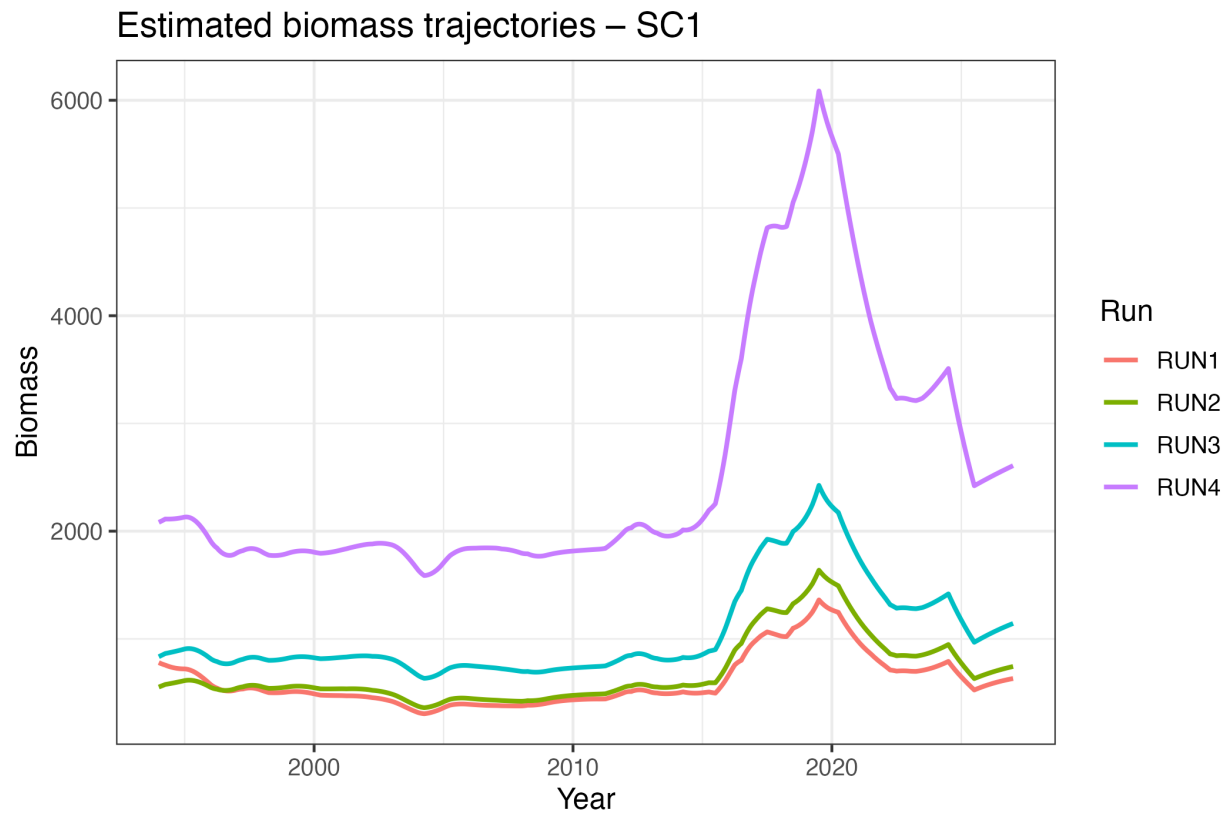


Figure 35: Estimated relative biomass trajectory for Scenario SC1 derived from the SPiCT model.

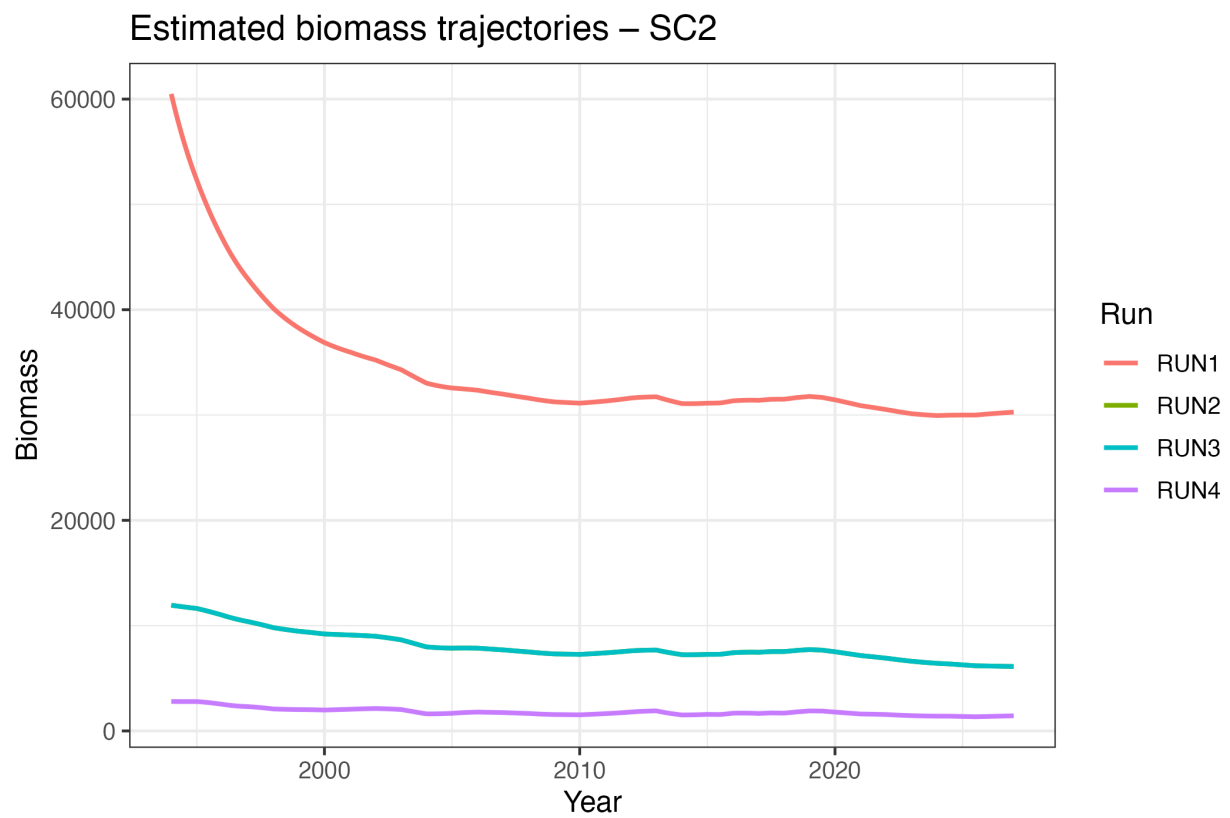


Figure 36: Estimated relative biomass trajectory for Scenario SC2 derived from the SPiCT model.

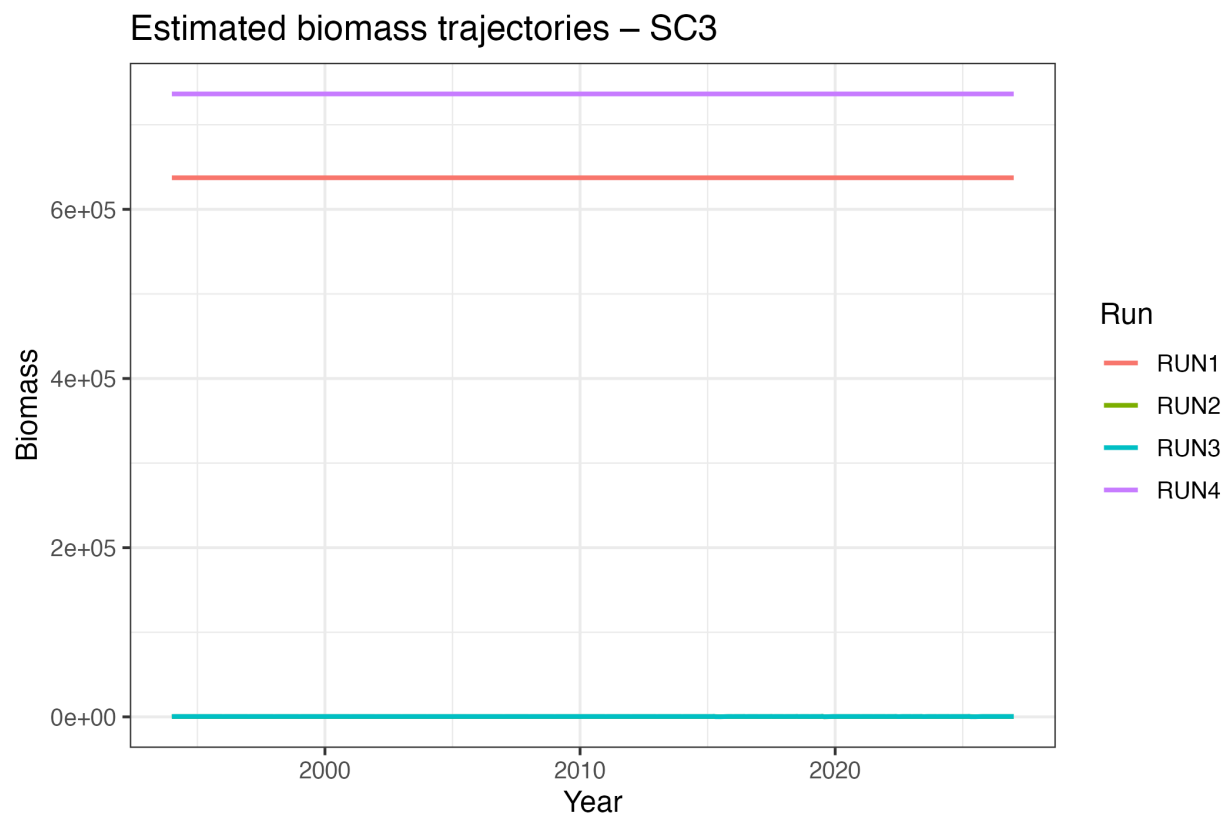


Figure 37: Estimated relative biomass trajectory for Scenario SC3 derived from the SPiCT model.

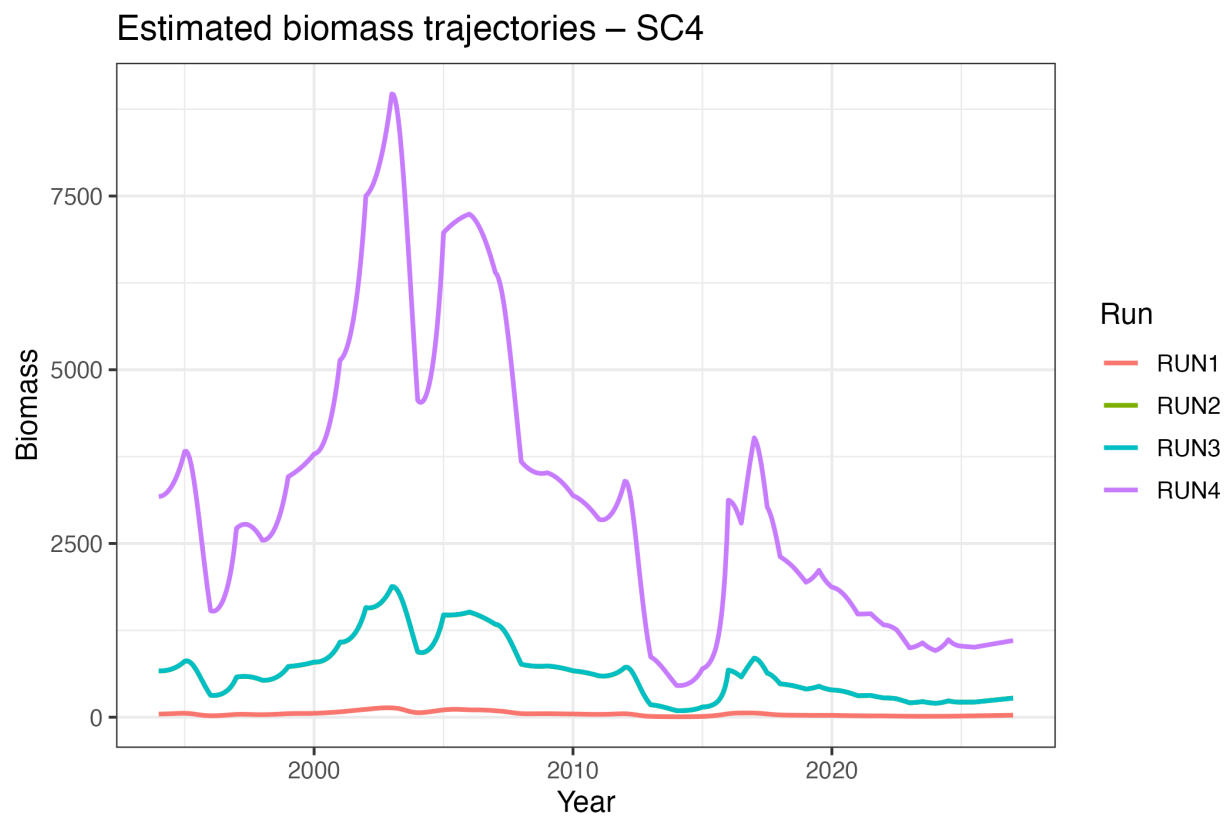


Figure 38: Estimated relative biomass trajectory for Scenario SC4 derived from the SPiCT model.

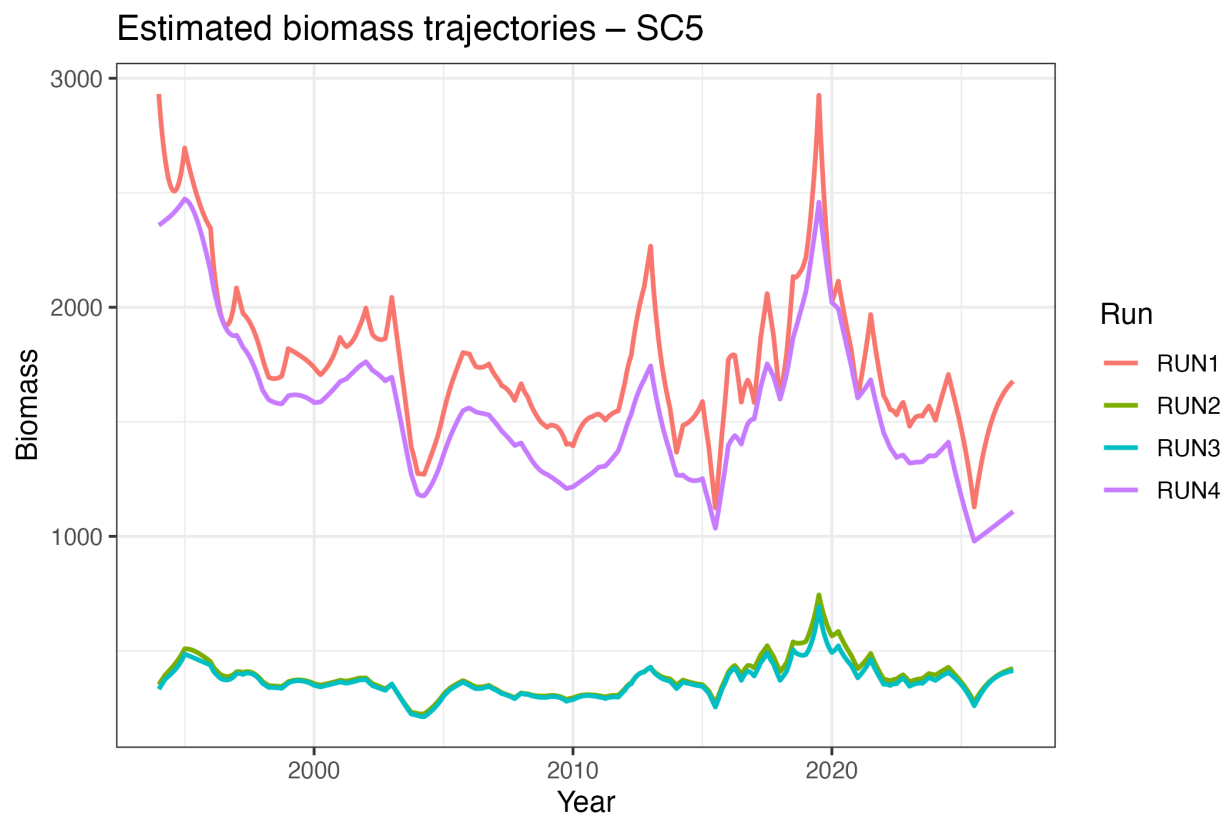


Figure 39: Estimated relative biomass trajectory for Scenario SC5 derived from the SPiCT model.

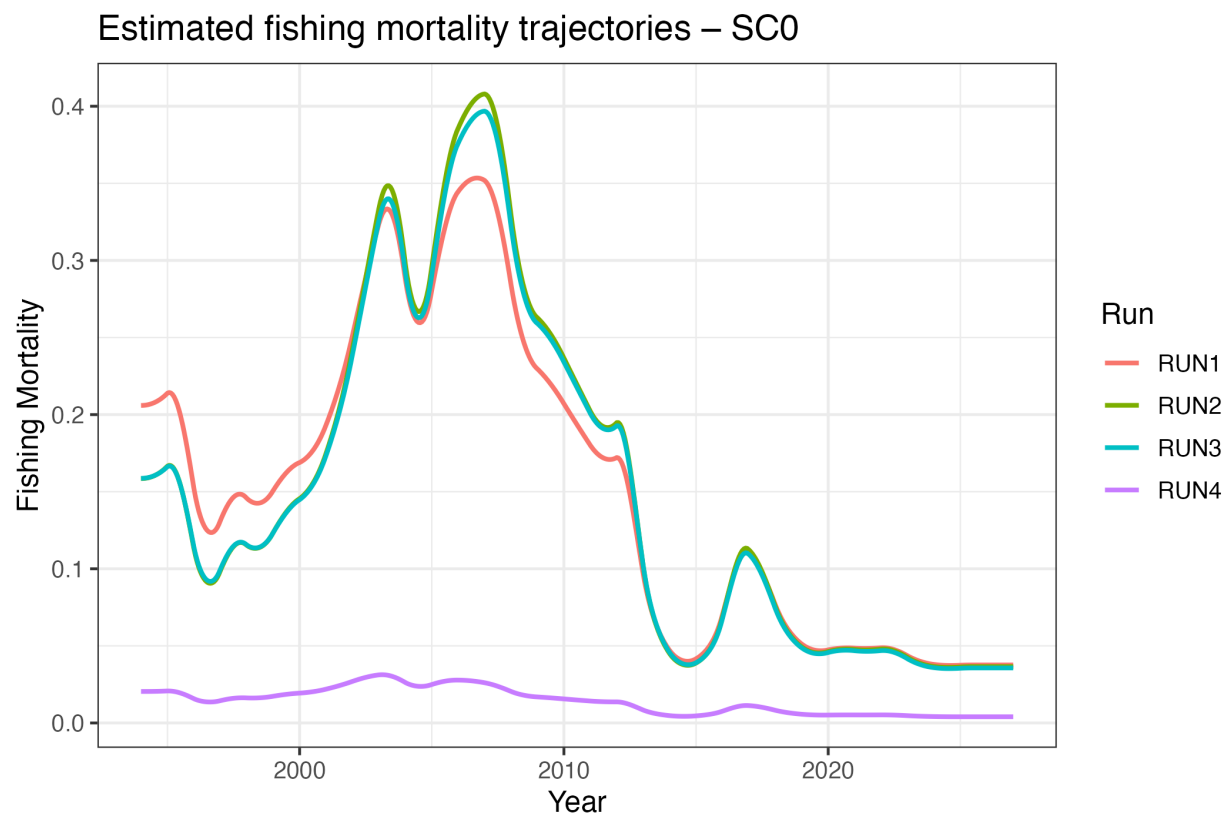


Figure 40: Estimated fishing mortality trajectory for Scenario SC0 derived from the SPiCT model.

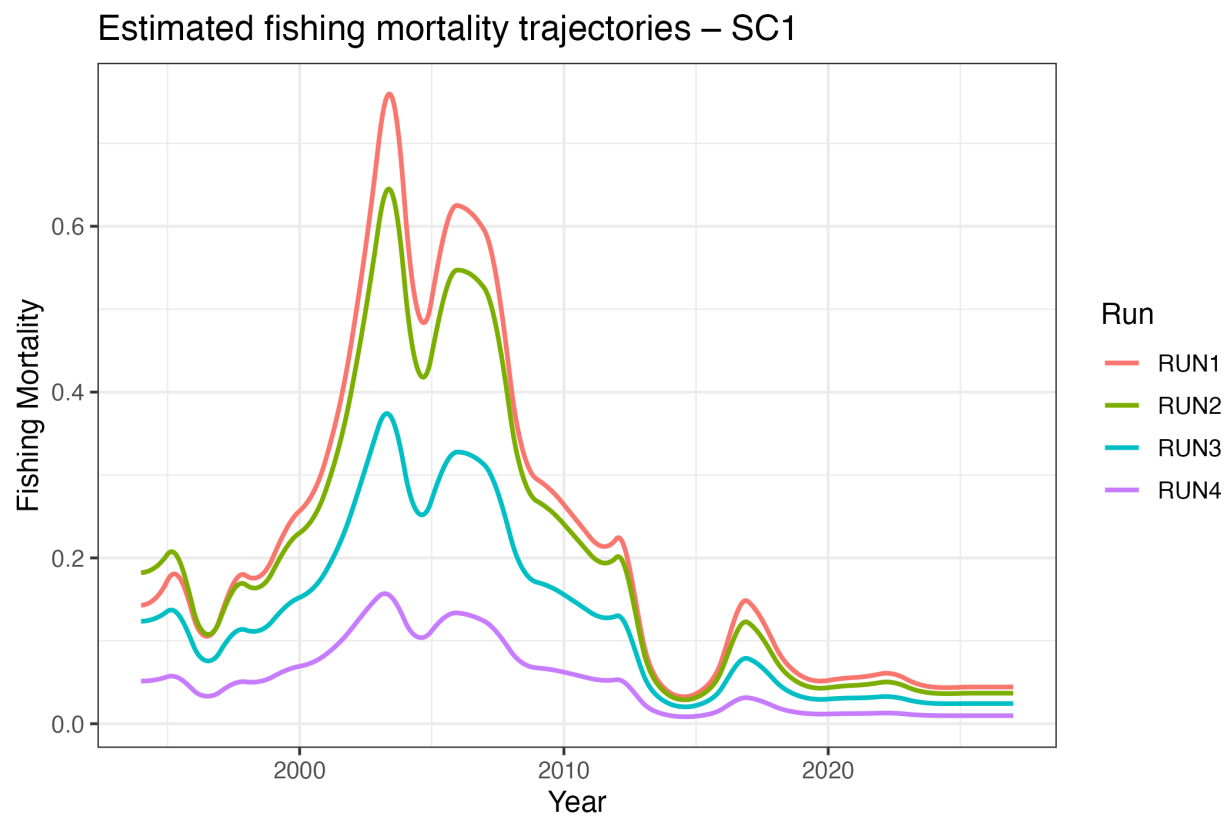


Figure 41: Estimated fishing mortality trajectory for Scenario SC1 derived from the SPiCT model.

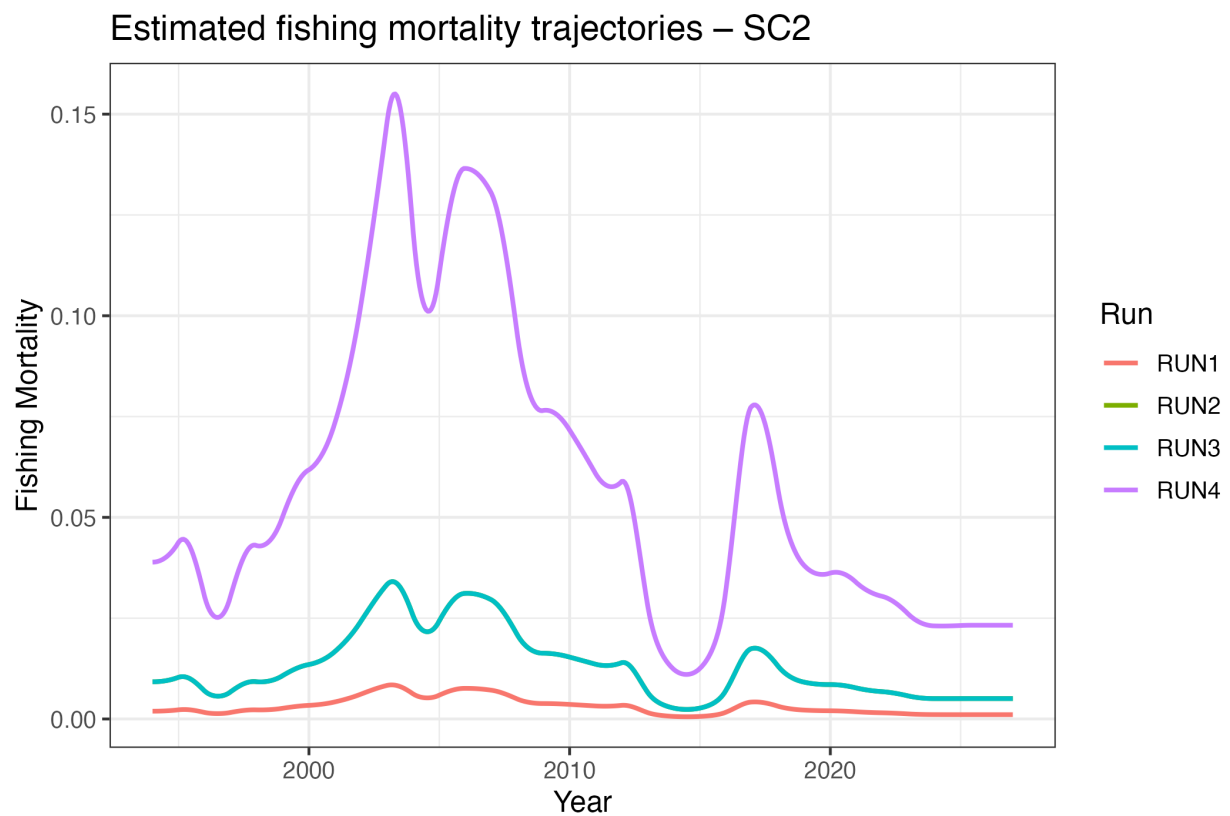


Figure 42: Estimated fishing mortality trajectory for Scenario SC2 derived from the SPiCT model.

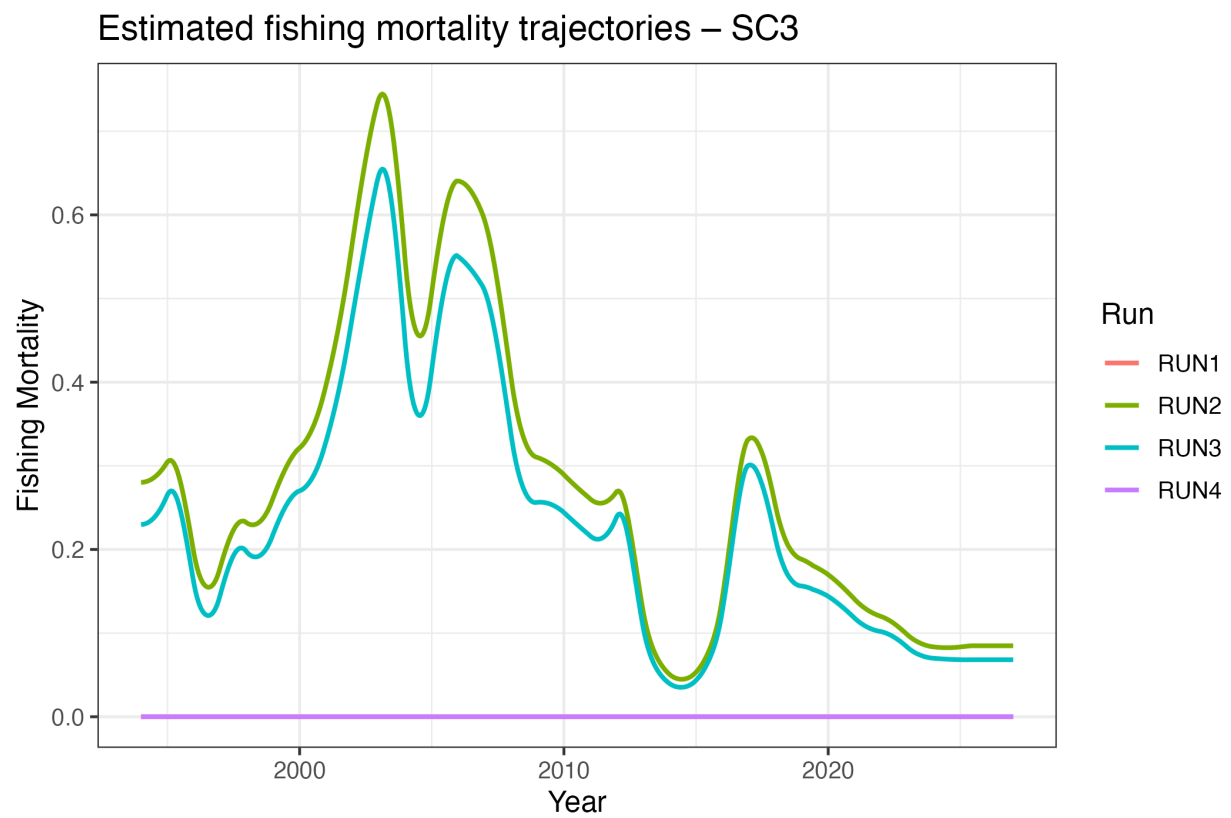


Figure 43: Estimated fishing mortality trajectory for Scenario SC3 derived from the SPiCT model.

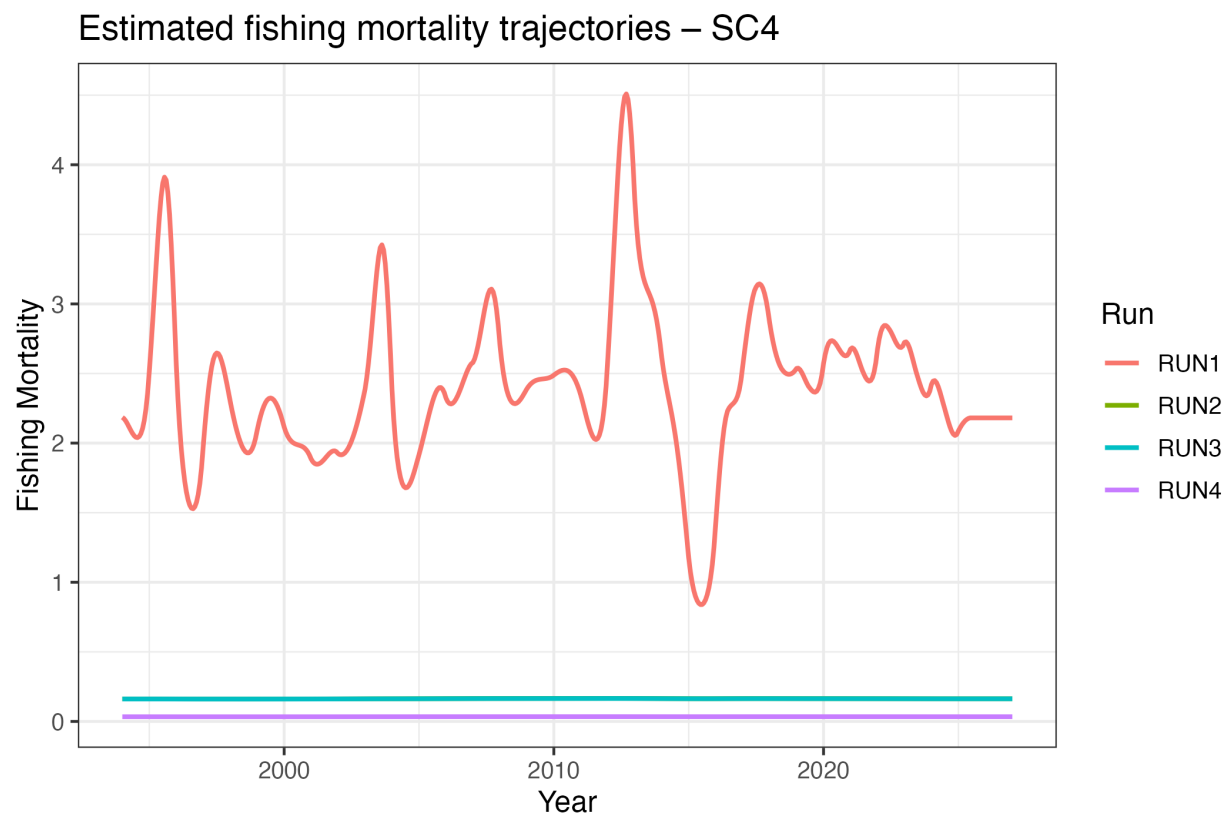


Figure 44: Estimated fishing mortality trajectory for Scenario SC4 derived from the SPiCT model.

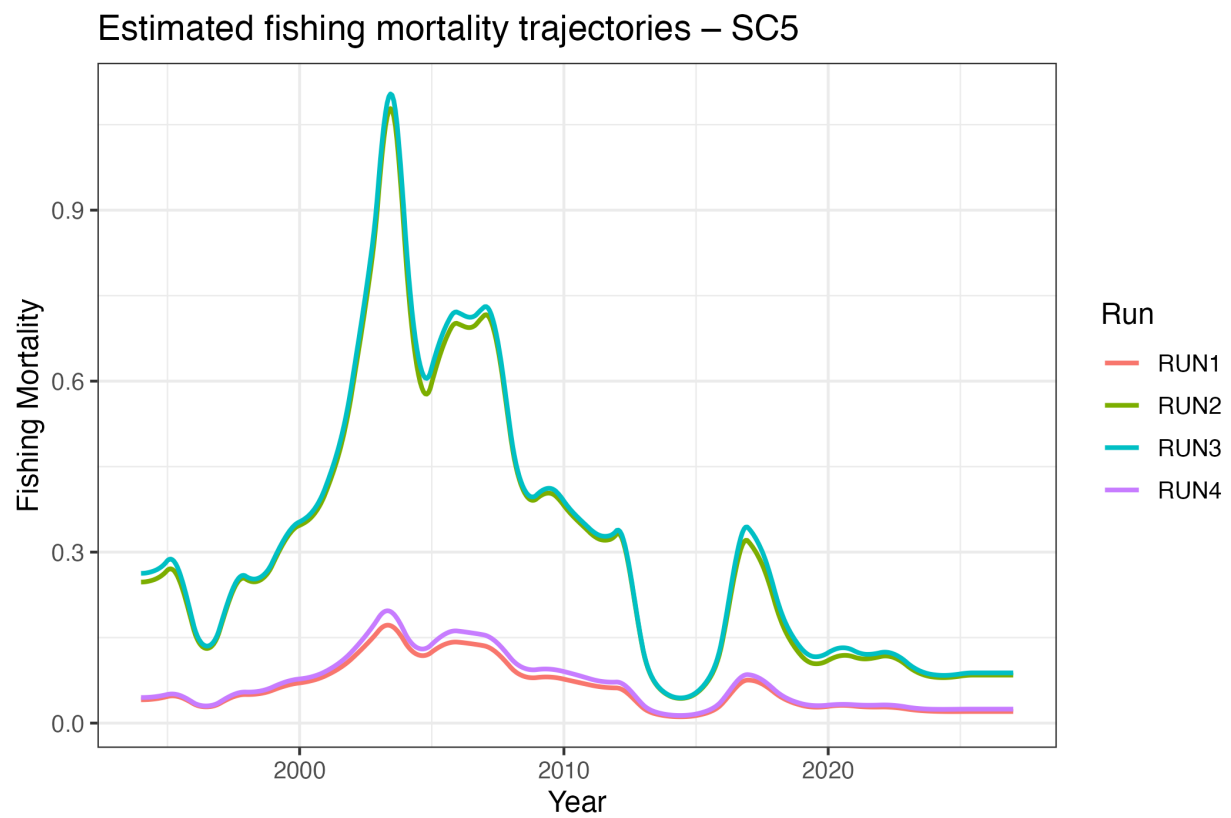


Figure 45: Estimated fishing mortality trajectory for Scenario SC5 derived from the SPiCT model.

Table 8: Summary of SPiCT model diagnostics and outputs by scenario and prior configuration (RUN).

Scenario	Run	Convergence	PDHess	AIC	Biomass_2025	FishingMortality_2025	ProdCurve	Mohn_BBmsy	Mohn_FFmsy
SC0	RUN1	FALSE	FALSE	81.74	11581.04	0.04	0.46	NA	NA
SC0	RUN2	FALSE	FALSE	75.16	11944.54	0.04	0.51	NA	NA
SC0	RUN3	FALSE	FALSE	79.84	12193.22	0.04	0.52	NA	NA
SC0	RUN4	TRUE	FALSE	85.69	105352.14	0.00	0.47	-0.22	1.87
SC1	RUN1	TRUE	FALSE	161.43	9751.92	0.04	0.23	-0.28	1.19
SC1	RUN2	TRUE	FALSE	156.39	11686.70	0.04	0.23	-0.08	0.19
SC1	RUN3	TRUE	FALSE	155.01	17693.59	0.02	0.44	-0.07	0.12
SC1	RUN4	TRUE	FALSE	154.64	43935.02	0.01	0.45	-0.07	0.15
SC2	RUN1	TRUE	FALSE	93.81	449899.34	0.00	0.48	0.02	0.87
SC2	RUN2	TRUE	FALSE	93.36	94100.83	0.01	0.50	NA	NA
SC2	RUN3	TRUE	FALSE	90.59	94100.95	0.01	0.50	NA	NA
SC2	RUN4	TRUE	FALSE	95.20	20637.30	0.02	0.59	0.12	-0.15
SC3	RUN1	FALSE	FALSE	-1101.78	9559775.29	0.00	0.09	NA	NA
SC3	RUN2	FALSE	FALSE	57.04	5780.24	0.08	0.68	NA	NA
SC3	RUN3	FALSE	FALSE	49.65	6403.39	0.07	0.74	NA	NA
SC3	RUN4	TRUE	FALSE	-874.48	11046283.28	0.00	0.99	0.00	0.00
SC4	RUN1	TRUE	FALSE	63.88	250.38	2.13	0.40	-0.31	0.12
SC4	RUN2	TRUE	FALSE	112.47	3286.40	0.16	0.49	0.00	-0.09
SC4	RUN3	TRUE	FALSE	109.71	3288.82	0.16	0.50	0.01	-0.10
SC4	RUN4	TRUE	FALSE	109.59	15493.14	0.03	0.45	0.07	-0.27
SC5	RUN1	TRUE	FALSE	316.98	21792.41	0.02	0.31	45.73	NA
SC5	RUN2	TRUE	FALSE	322.34	5403.90	0.08	0.30	0.83	NA
SC5	RUN3	TRUE	FALSE	320.42	5194.90	0.09	0.45	NA	0.01
SC5	RUN4	TRUE	FALSE	317.43	17709.07	0.02	0.51	0.01	-0.07

References

- Acom, I. C. M. (2015). *ICES WKLIFE V REPORT 2015 Methodologies based on Life-history Traits , Development of Quantitative Assessment Exploitation Characteristics and other Report of the Fifth Workshop on the Relevant Parameters for Data-limited Stocks (WKLIFE V) Lisbon , Port.* (October), 5–9.
- Bell, E. (2019). Separable length cohort method (SCA). *Length-Based Reference Point Estimation*.
- Cousido-Rocha, M., Cerviño, S., Alonso-Fernández, A., Gil, J., González-Herraiz, I., Rincón, M. M., ... Pennino, M. G. (2022). Applying length-based assessment methods to fishery resources in the bay of biscay and iberian coast ecoregion: Stock status and parameter sensitivity. *Fisheries Research*, 248, 106197. <https://doi.org/10.1016/j.fishres.2021.106197>
- Dobby, H. (2019). *Nepref: Calculates per-recruit reference points for nephrops*.
- European Union. (2023). *Regulation (EU) 2023/194 fixing fishing opportunities for 2023*.
- European Union. (2024). *Regulation (EU) 2024/225 establishing a fisheries closure for norway lobster in FU 30*.
- González Herraiz, I., Vila, Y., Cardinale, M., Berg, C. W., Winker, H., Azevedo, M., ... Pennino, M. G. (2023). First maximum sustainable yield advice for the nephrops norvegicus stocks of the northwest iberian coast using stochastic surplus production model in continuous time (SPiCT). *Frontiers in Marine Science*, 10, 1062078. <https://doi.org/10.3389/fmars.2023.1062078>
- ICES. (2009). *Report of the benchmark workshop on nephrops (WKNEP), 2–6 march 2009, aberdeen, UK* (No. CM 2009/ACOM:33). International Council for the Exploration of the Sea.
- ICES. (2017). *Report of the benchmark workshop on nephrops stocks (WKNEP)* (No. CM 2016/ACOM:38). ICES.
- ICES. (2020). *Workshop on methodologies for nephrops reference points (WKNephrops)* (No. 3; Vol. 2). ICES Scientific Reports. <https://doi.org/10.17895/ices.pub.5981>
- ICES. (2021a). *Benchmark workshop on the development of MSY advice for category 3 stocks using SPiCT (WKMSYSPICT)* (No. 20; Vol. 3). ICES Scientific Reports. <https://doi.org/10.17895/ices.pub.7919>
- ICES. (2021b). *Working group for the bay of biscay and the iberian waters ecoregion (WGBIE)* (No. 48; Vol. 3). ICES Scientific Reports. <https://doi.org/10.17895/ices.pub.8212>
- ICES. (2022). *Working group for the bay of biscay and the iberian waters ecoregion (WGBIE)* (No. 52; Vol. 4). ICES Scientific Reports. <https://doi.org/10.17895/ices.pub.20068988>
- ICES. (2023a). *Advice on fishing opportunities*. International Council for the Exploration of the Sea. <https://doi.org/10.17895/ices.advice.22240624>

- ICES. (2023b). *ICES guidance for completing single-stock advice 2023* (p. 64). International Council for the Exploration of the Sea.
- Leocádio, A., Weetman, A., & Wieland, K. (Eds.). (2018). *Using UWTV surveys to assess and advise on nephrops stocks* (No. 340; p. 49). International Council for the Exploration of the Sea. <https://doi.org/10.17895/ices.pub.4370>
- Mildenberger, T. K., Kokkalis, A., & Berg, C. W. (2023). *Guidelines for the stochastic production model in continuous time (SPiCT)*. Retrieved from https://github.com/DTUAqua/spict/raw/master/spict/inst/doc/spict_guidelines.pdf
- Payá, I., Cristian, C. R. D. B., Andrades, M. C., Faúndez, F. C. M. E. L., Rubio, R. T. O. A. Y., & Basualto, M. J. Z. (2014). *INFORME FINAL Convenio II: Estatus y posibilidades de explotación biológicamente sustentables de los principales recursos pesqueros nacionales 2014 Proyecto 2.16: Revisión de los puntos biológicos de referencia (Rendimiento Máximo Sostenible) en las pesqu* (p. 855). Retrieved from https://www.estadonacion.or.cr/files/biblioteca%7B/_%7Dvirtual/021/politica/Alvarado%7B/_%7DConflicto%7B/_%7DMoin
- Vila, Y., & Burgos, C. (2022). *New area proposed for the ISUNEPCA UWTV survey in the gulf of cadiz (FU 30)* (No. 52; p. 847). International Council for the Exploration of the Sea. <https://doi.org/10.17895/ices.pub.20068988>
- Vila, Y., Burgos, C., & Soriano, M. M. (2016). *Nephrops (FU 30) UWTV survey on the gulf of cadiz grounds*. International Council for the Exploration of the Sea.
- Vila, Y., & Quinzán, M. (2024). *Stochastic surplus production model in continuous time (SPiCT) explorations in the nephrops FU30 stock (gulf of cadiz)* (Technical Report No. WGBIE 2024). Lisbon, Portugal: Instituto Español de Oceanografía (IEO-CSIC), Centro Oceanográfico de Cádiz.